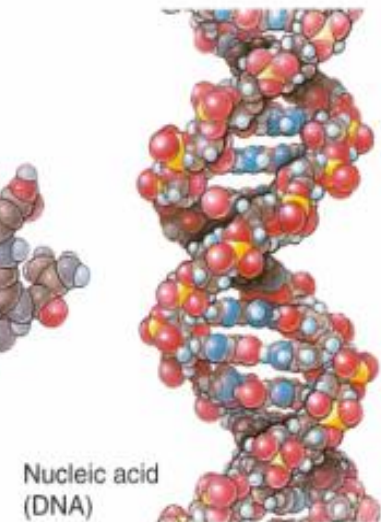
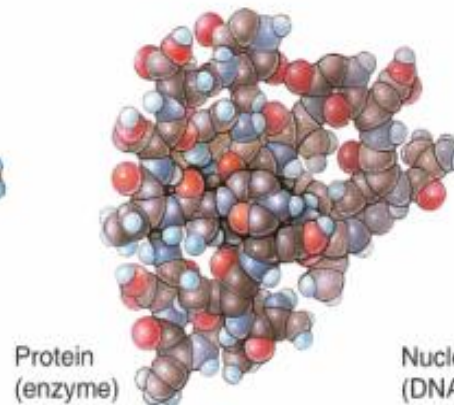
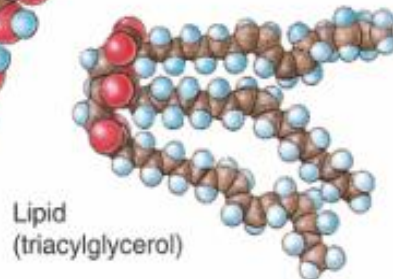
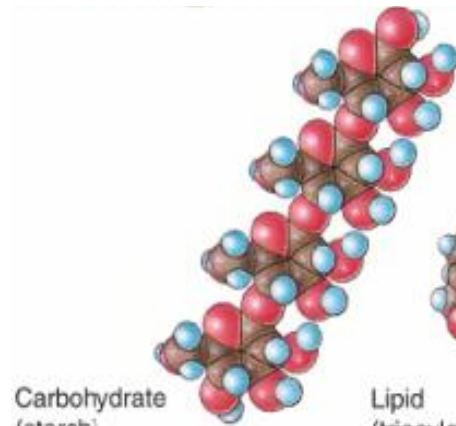
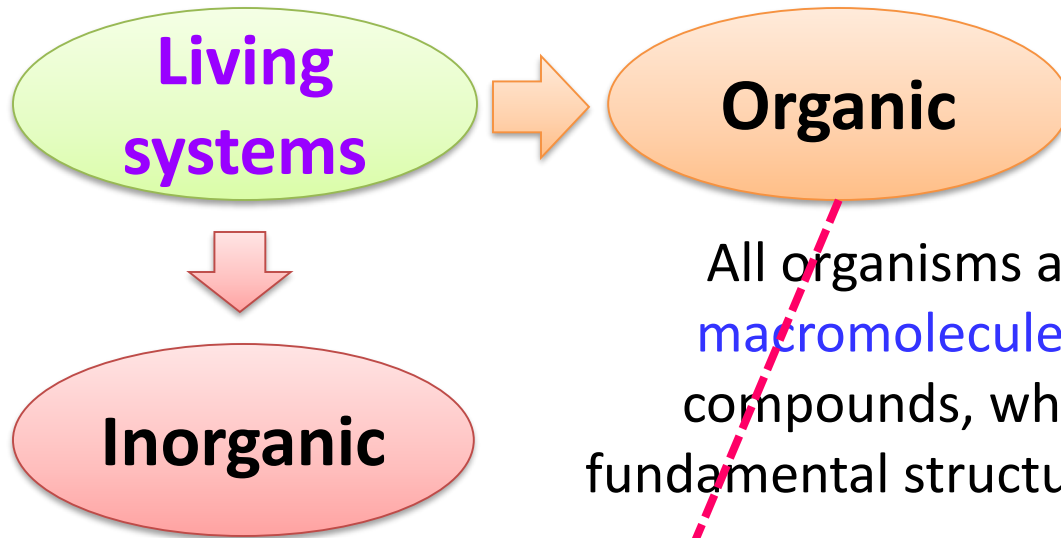


Biological Macromolecules

Lecture 6-9 MRIs



Biological Macromolecules



All organisms are composed of **biological macromolecules**. These are carbon-based compounds, which means that they have a fundamental structure composed of carbon atoms.

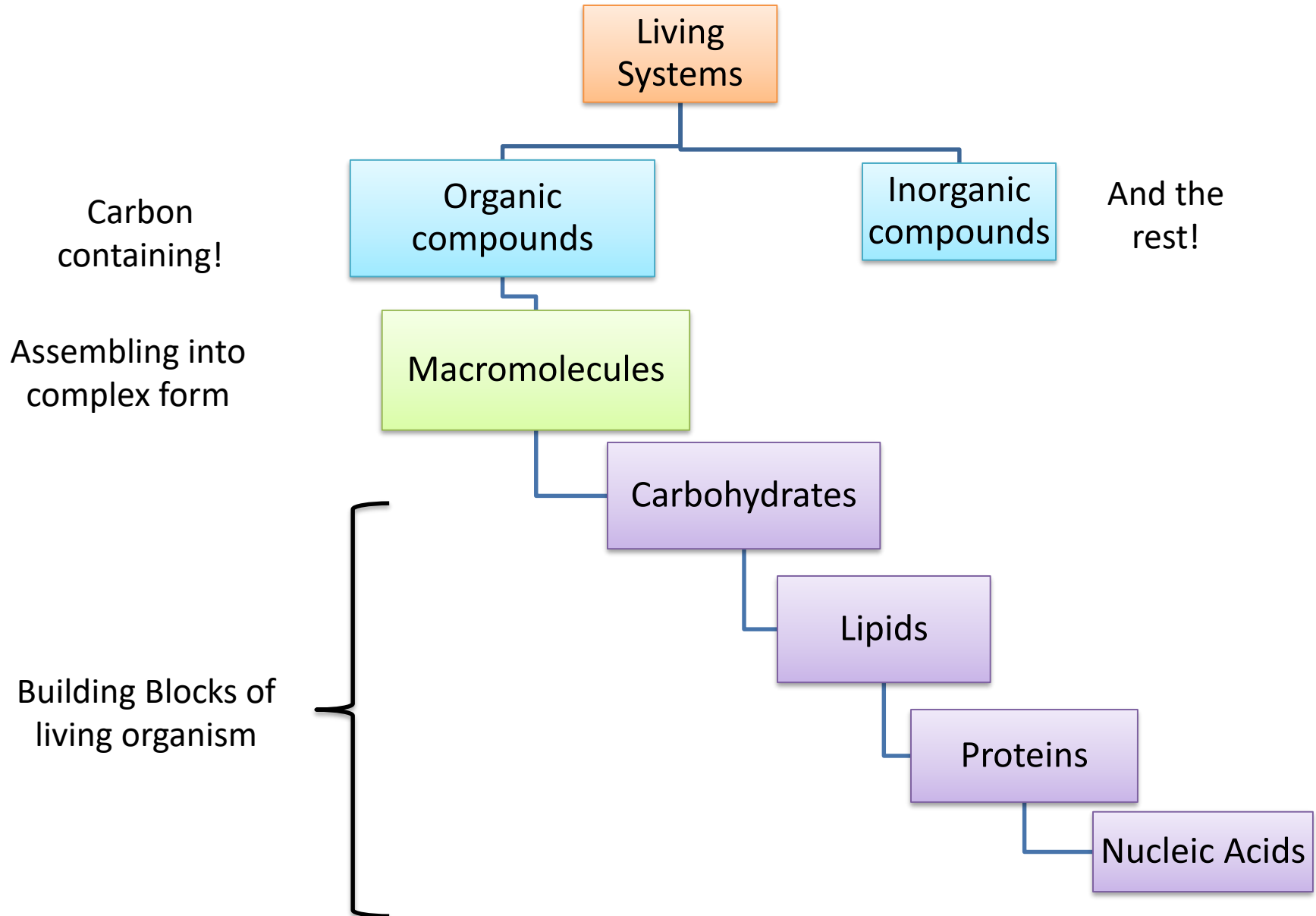
Macromolecules are very large molecules that are formed by the polymerization of smaller molecules called monomers.

There are **four major classes** of biological macromolecules:

1. Carbohydrates
2. Proteins
3. Nucleic acids
4. Lipids

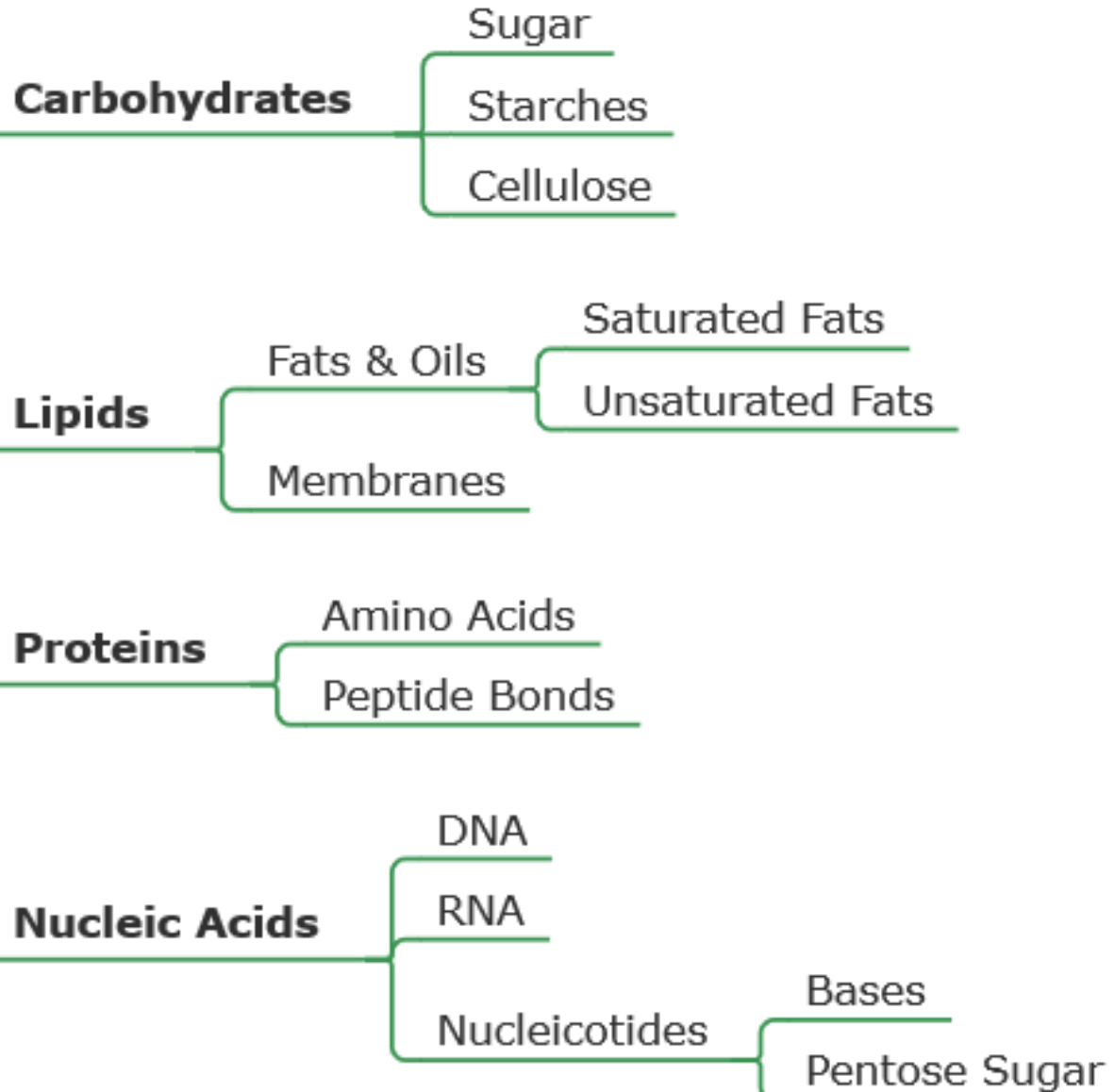
	Cellular composition
– Carbohydrates	15%
– Proteins	50%
– Nucleic acids	25%
– Lipids	10%

Introduction



Biological Macromolecules

Biomolecules

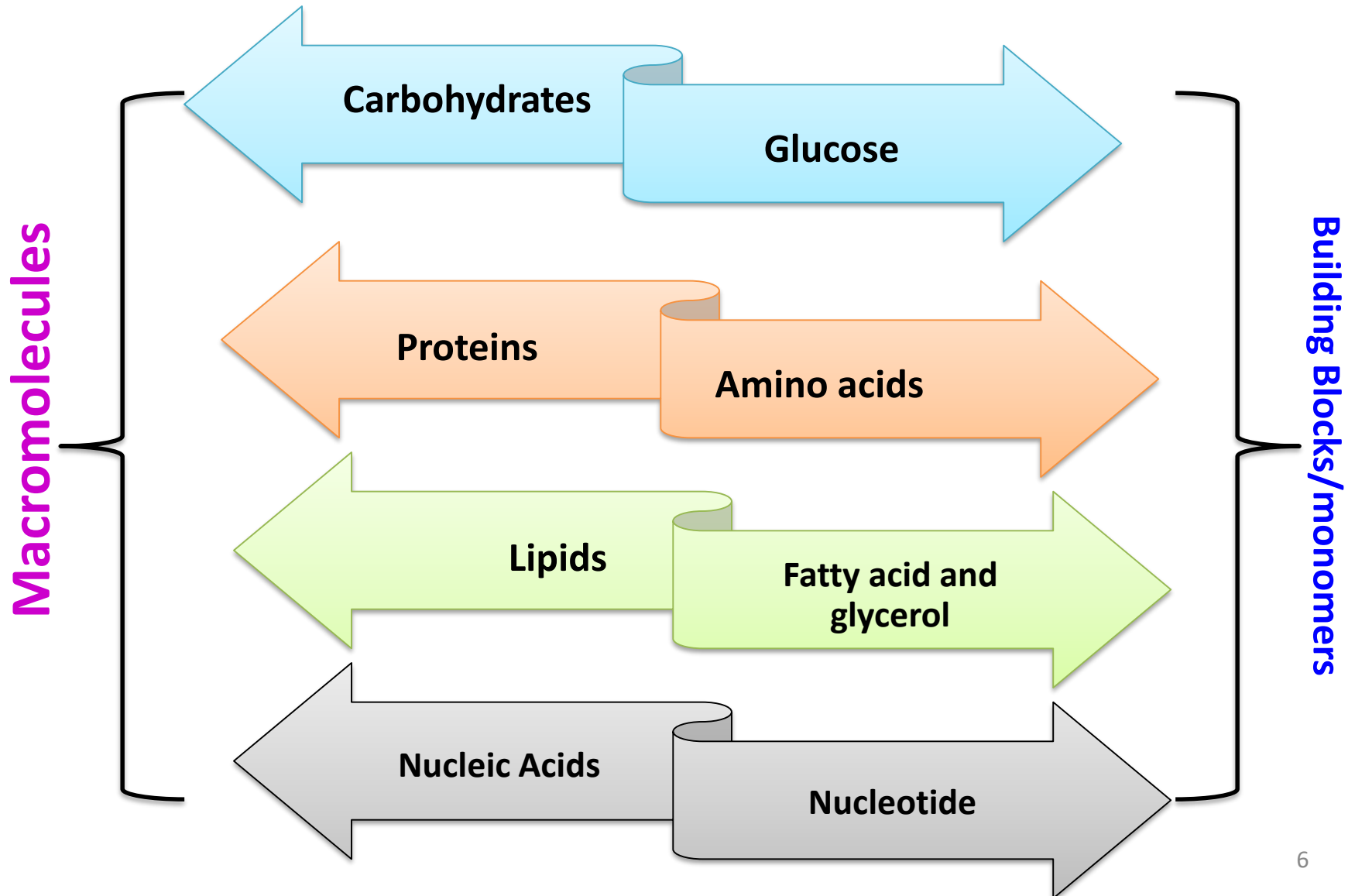


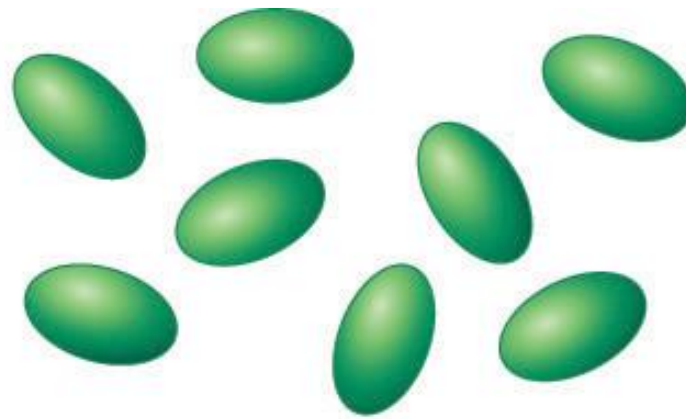
How are macromolecules formed?

Monomers are small molecules or building blocks that may be joined together in a repeating fashion to form more complex molecules called polymers.

A **polymer** may be a natural or synthetic macromolecule comprised of repeating units of a smaller molecule (monomers).

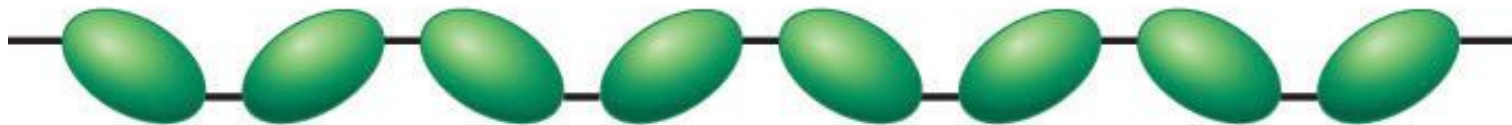
Monomers and macromolecules





Monomers

Polymerization



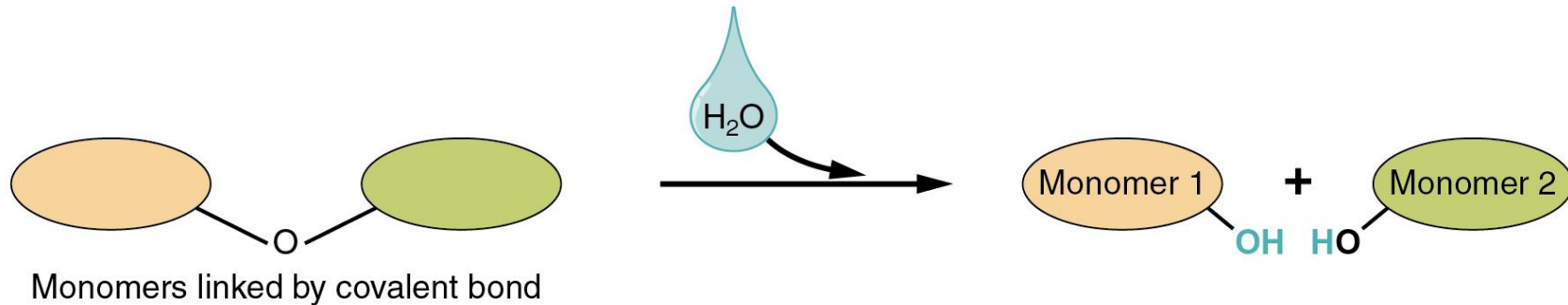
Polymer

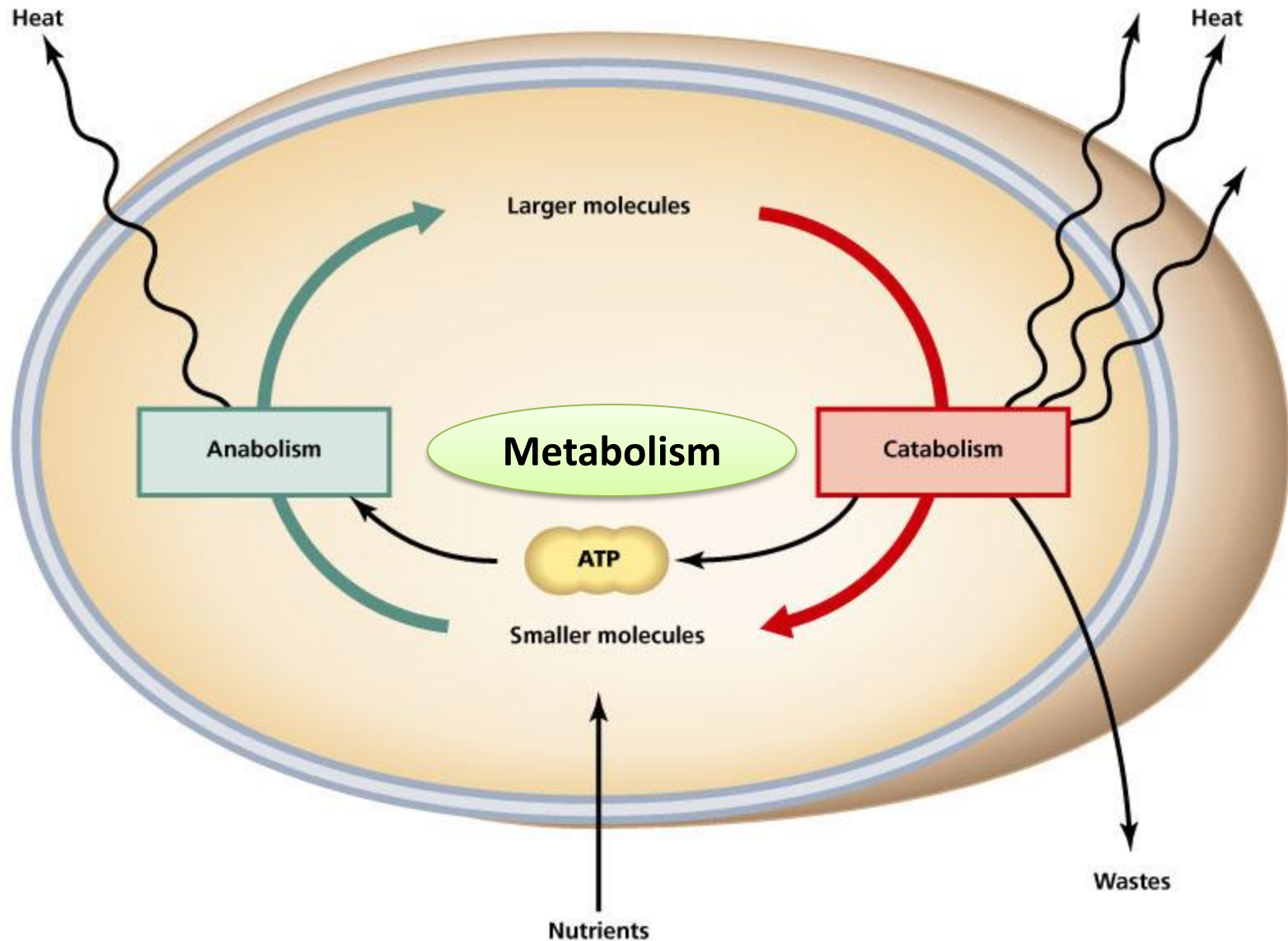
How are macromolecules broken or digested?

Hydrolysis

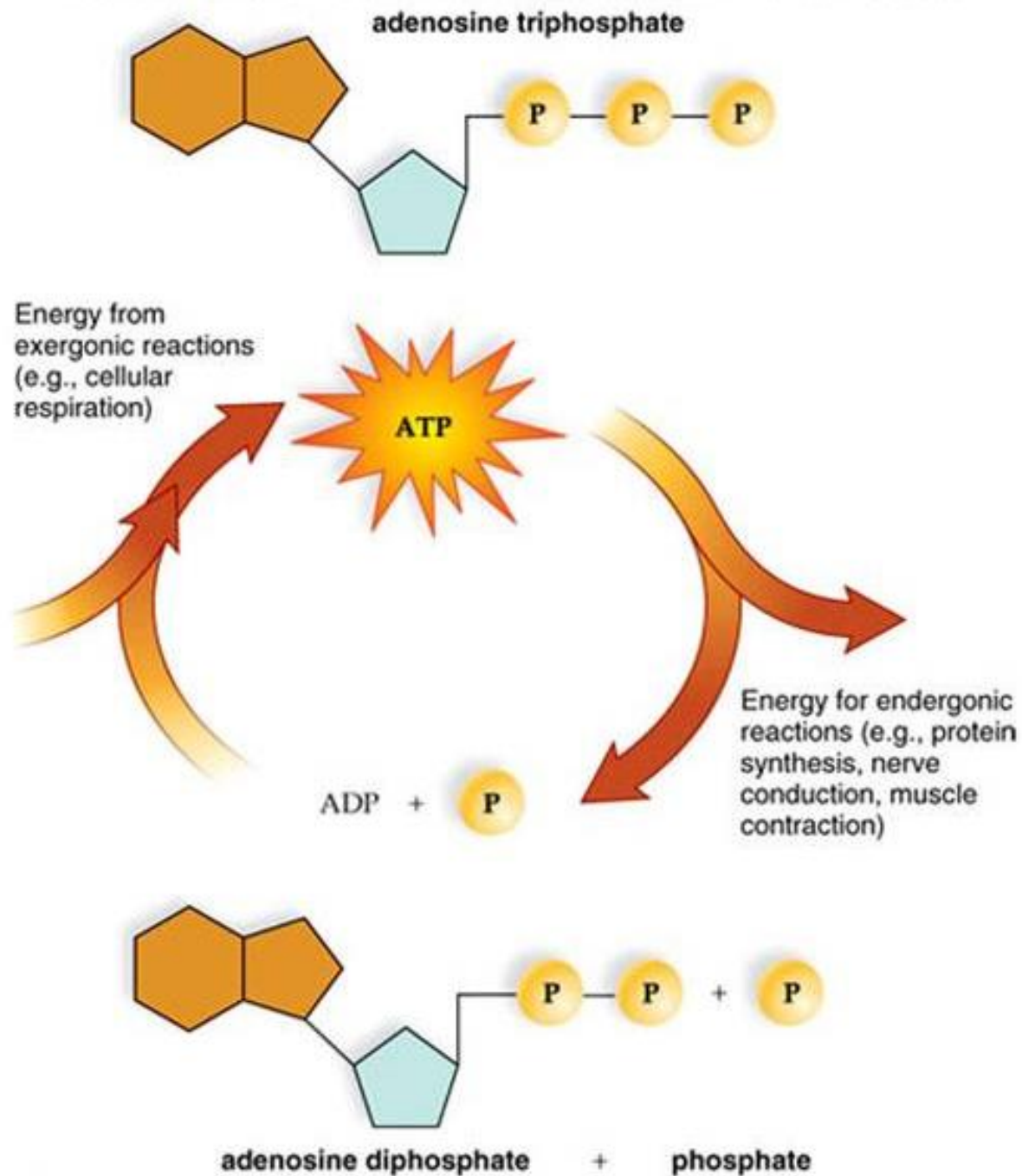
Hydrolysis

Monomers are released by the addition of a water molecule, adding OH to one monomer and H to the other.





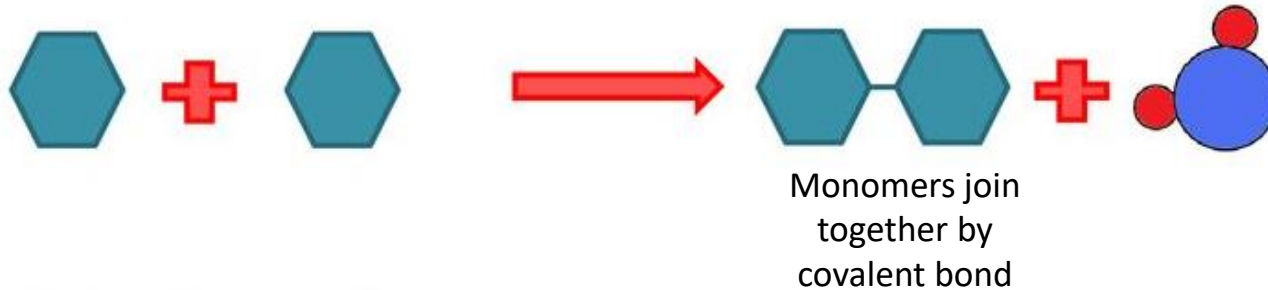
Metabolism refers to activities by which cells acquire and use energy as they make and break apart organic compounds.



a.

- **Condensation reaction:**

Removing water from two **monomers** to form a **dimer**



- **Hydrolysis reaction:**

Adding water can break **polymers** and **dimers** back into **monomers**



A **condensation reaction**, also commonly referred to as **dehydration synthesis**, is a chemical reaction in which two molecules or moieties (functional groups) combine to form a larger molecule, together with the loss of a small molecule (usually **water in the biological system**).

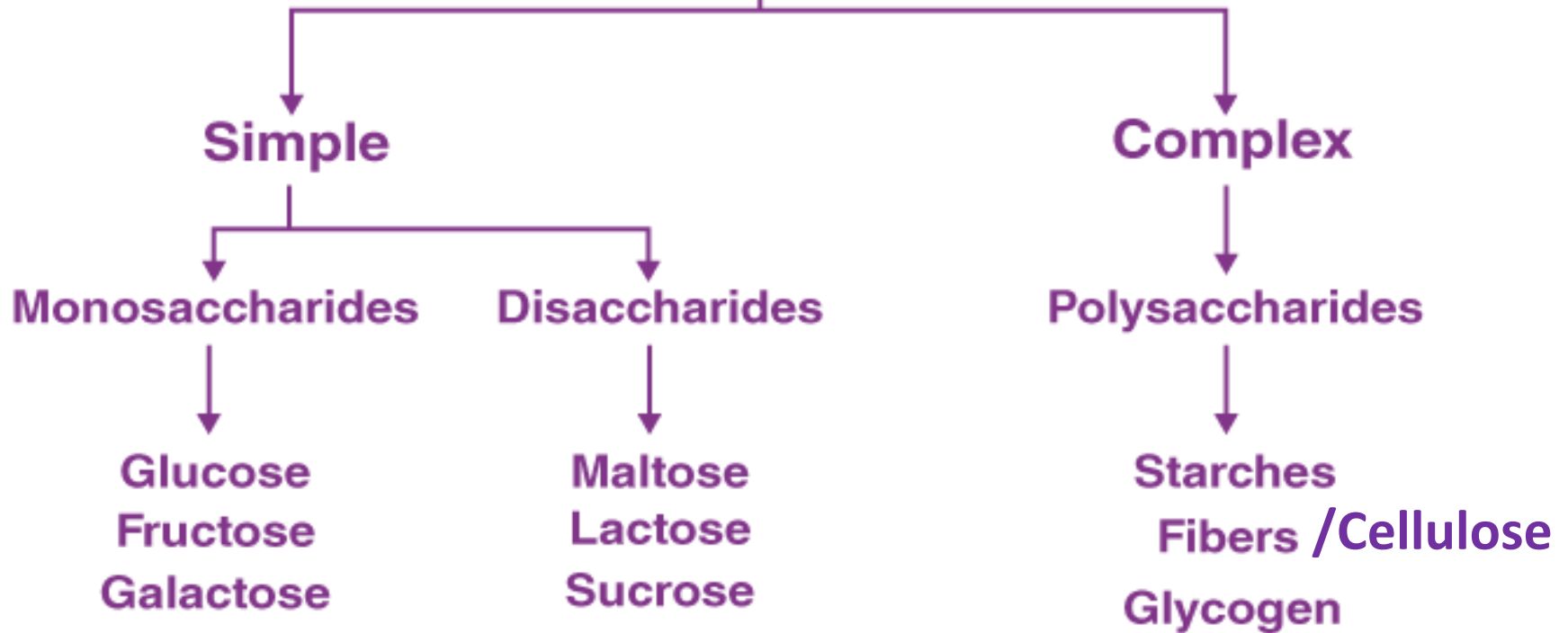
Learning About Carbohydrates



Carbohydrate

- The hydrate of carbon is known as carbohydrates.
- Chemically carbohydrates are polyhydroxy aldehydes or ketones
- Molecules with a 1:2:1 ratio of carbon, hydrogen, oxygen
- Common symbol: $(CH_2O)_n$
- C—H covalent bonds hold much energy
 - Carbohydrates are good energy storage molecules
 - Examples: sugars, starch, glucose

Carbohydrates



Monosaccharide

Fruits, vegetables,
honey, nuts



Disaccharide-

Sugar, milk



Polysaccharides-

Rice, potato,
wheat, corn

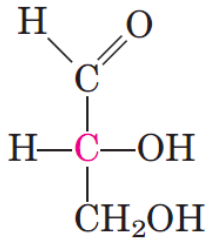
Functions of Carbohydrates

- Providing energy and regulation
- Store energy as glycogen
- Build macromolecules
- Sparing the use of proteins and fats for energy
- Preventing ketosis and breakdown of fatty acids and
- Biological recognition processes
- Flavor and Sweeteners
- Dietary fiber, which is also a form of carbohydrate, is essential for the elimination of waste materials and toxins from the body

Monosaccharides

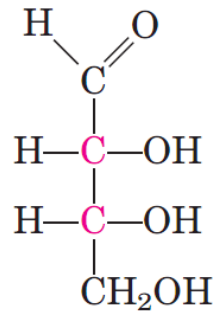
Monosaccharides: Examples of Aldoses

Three carbons



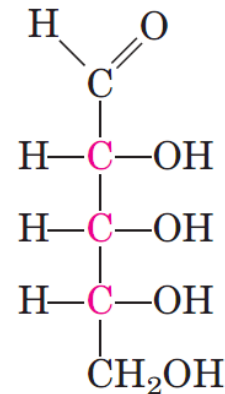
D-Glyceraldehyde

Four carbons



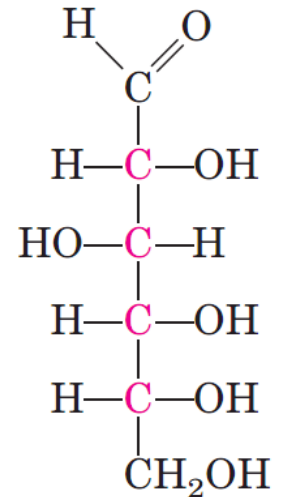
D-Erythrose

Five carbons



D-Ribose

Six carbons



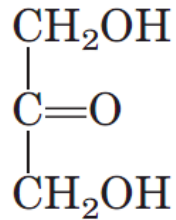
D-Glucose

**Main “fuel” for
bacteria, plants
and animal cells**

Monosaccharides

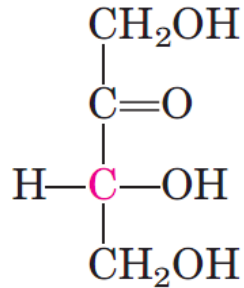
Monosaccharides: Examples of Ketoses

Three carbons



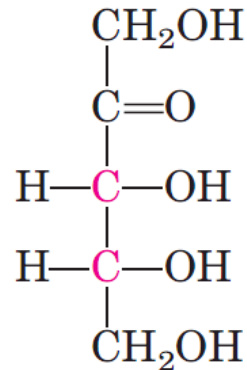
Dihydroxyacetone

Four carbons



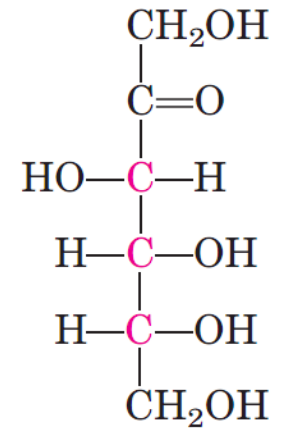
D-Erythrulose

Five carbons



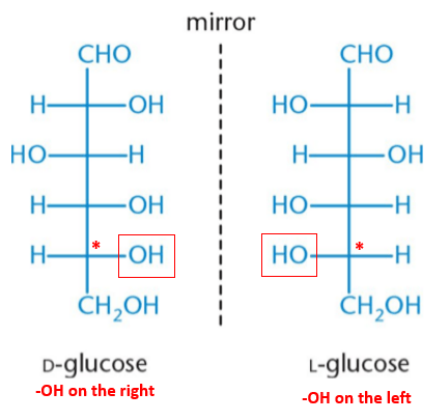
D-Ribulose

Six carbons

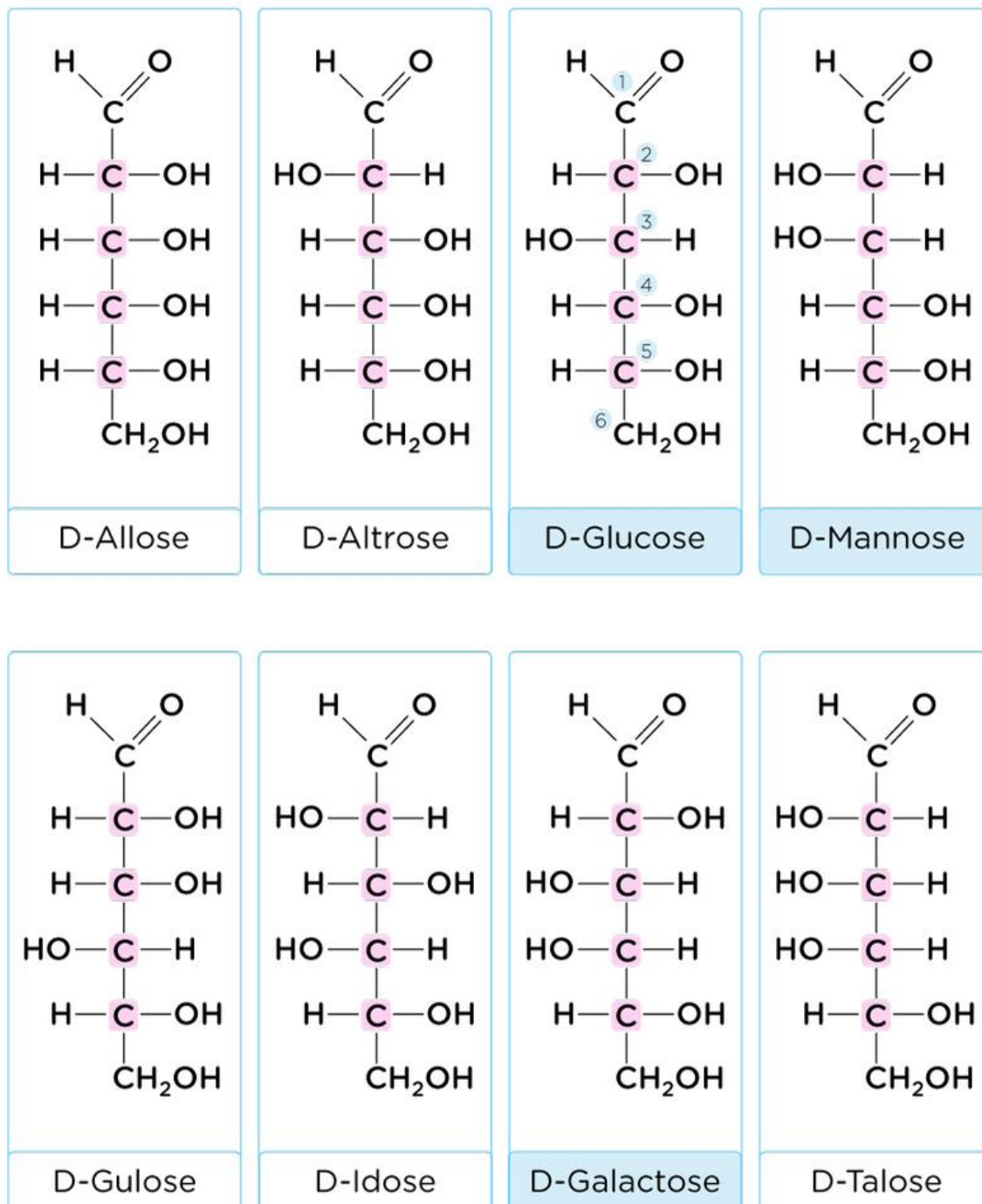


D-Fructose

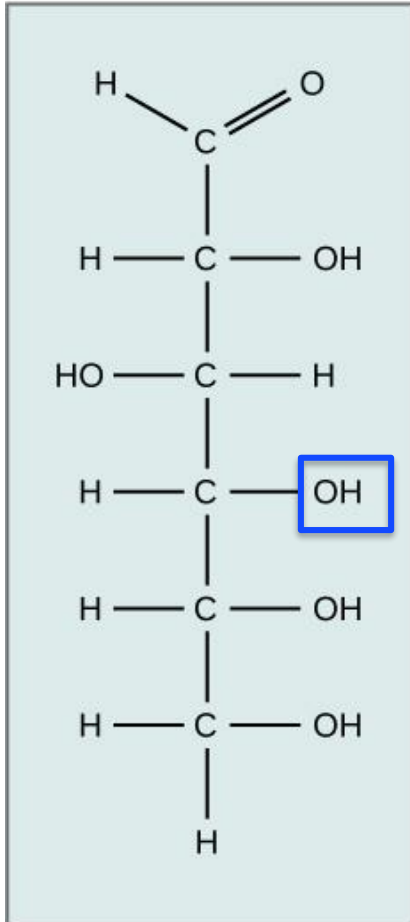
Epimers of D-glucose



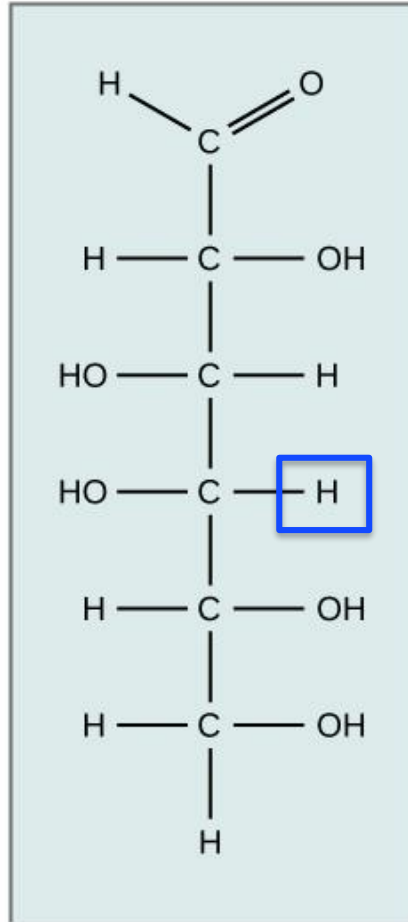
* chiral carbon



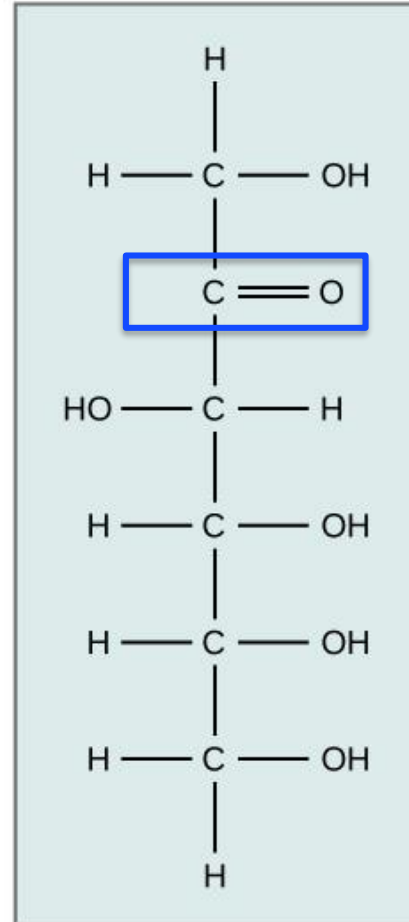
Glucose



Galactose

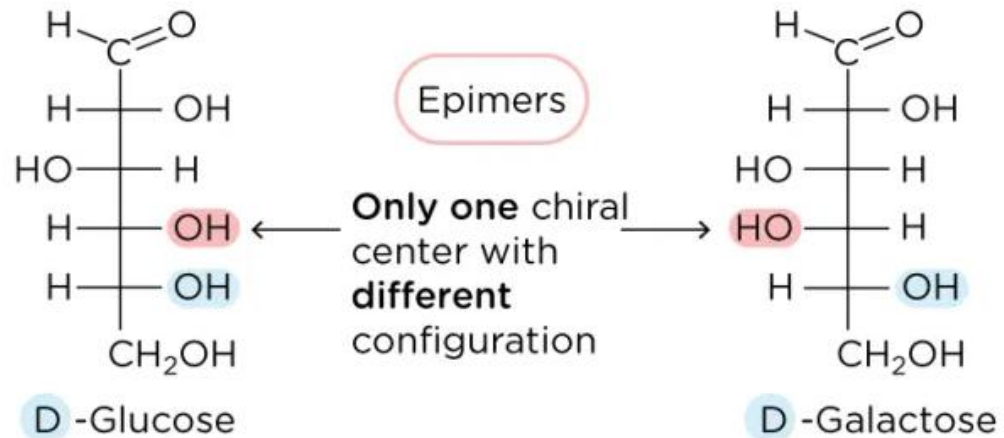


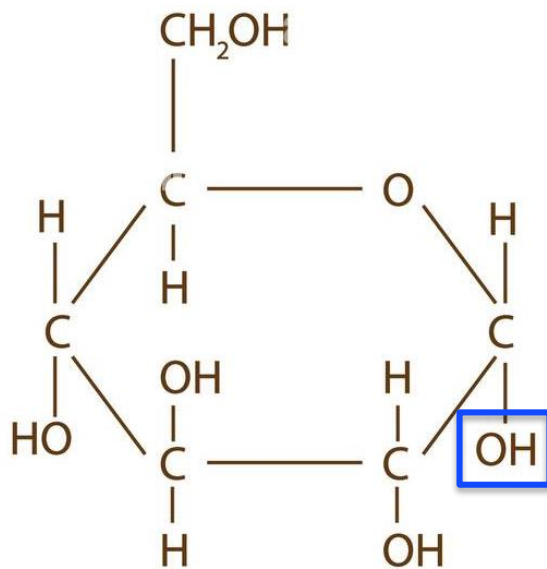
Fructose



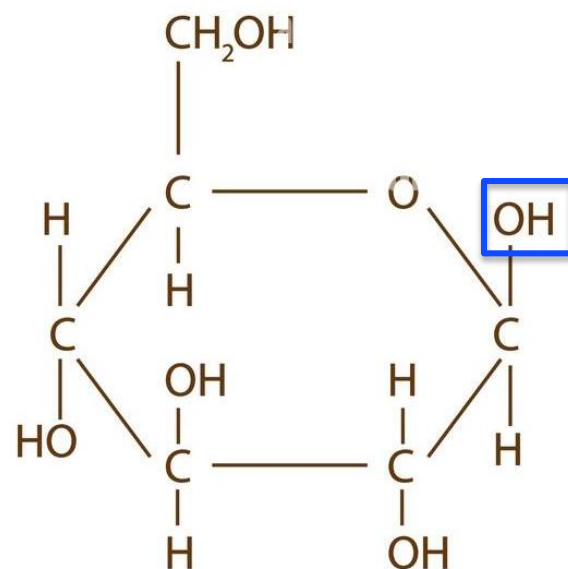
Glucose, galactose, and fructose are all hexoses. They are structural isomers, meaning they have the same chemical formula ($C_6H_{12}O_6$) but a different arrangement of atoms.

D-glucose and D-galactose are epimeric at carbon-4

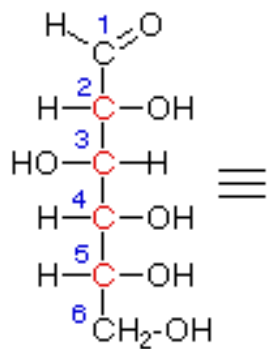




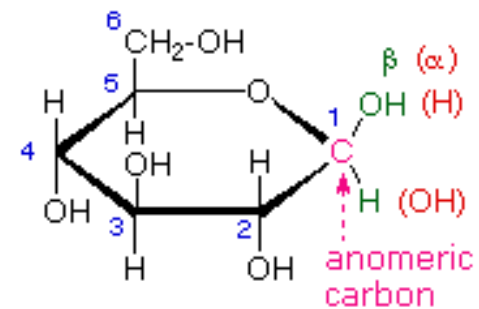
α -D-Glucose



β -D-Glucose

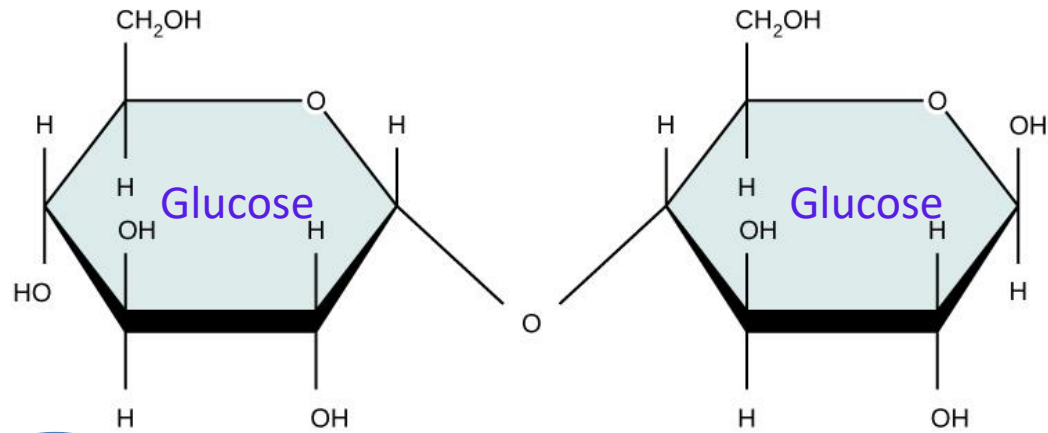
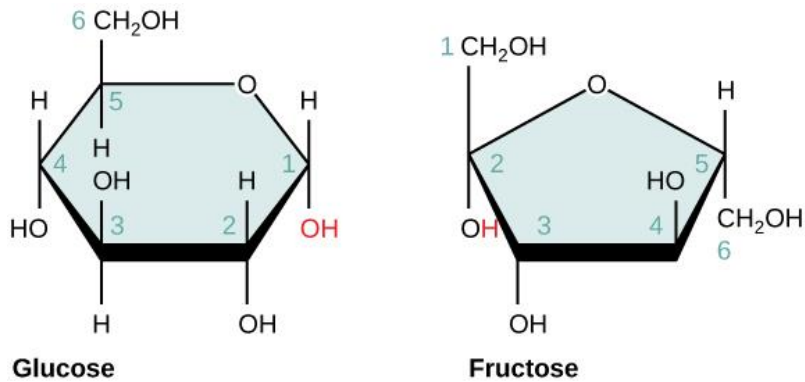


D-(+)-glucose



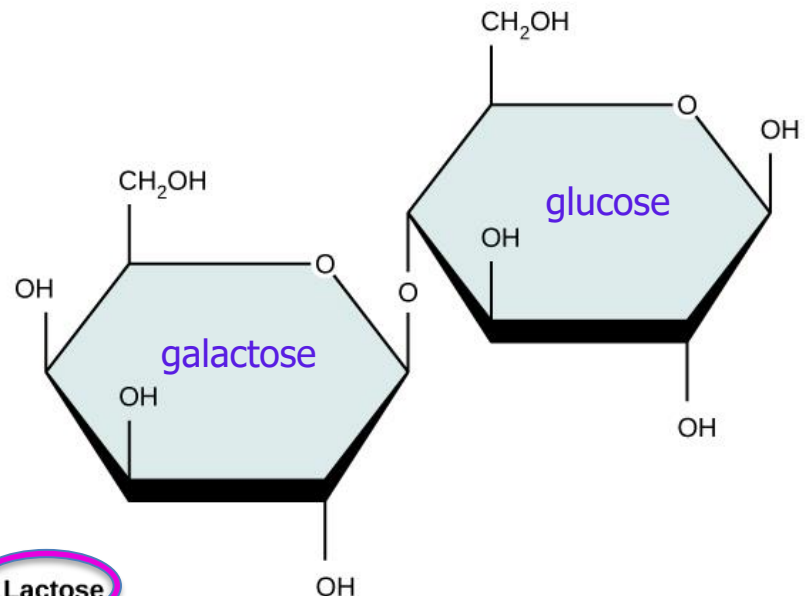
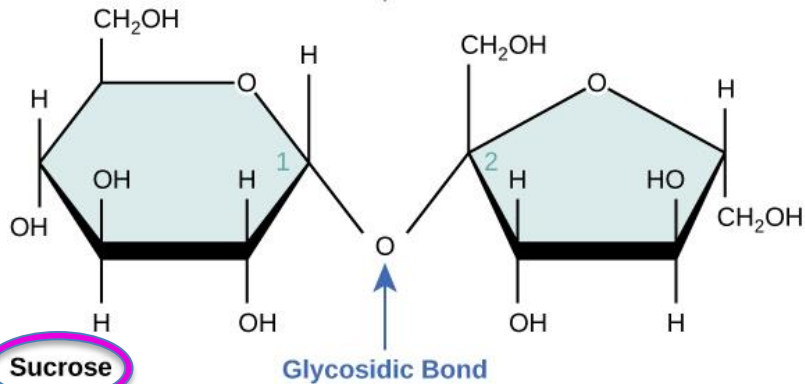
D-glucopyranose

Disaccharides



Maltose

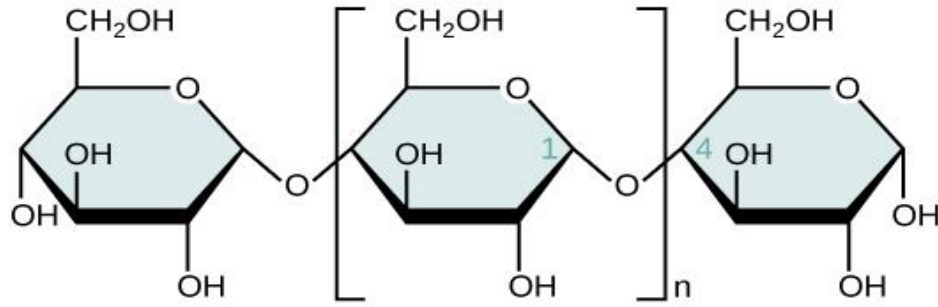
Maltose (grain sugar) = glucose + glucose



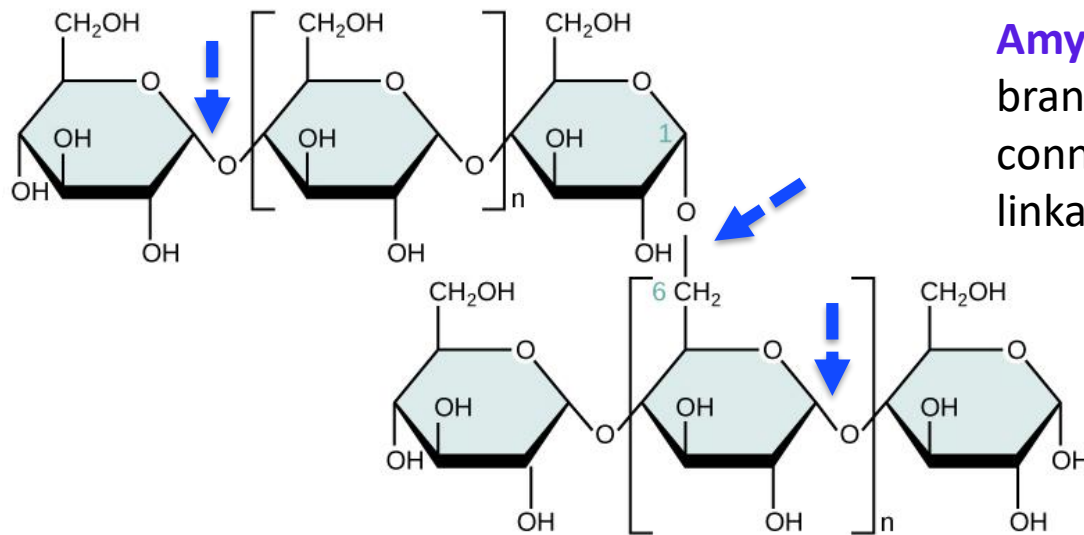
Lactose

(milk sugar) = glucose + galactose

Polysaccharides



Amylose (20-25%) is composed of unbranched chains of glucose monomers connected by α -1,4 glycosidic linkages.



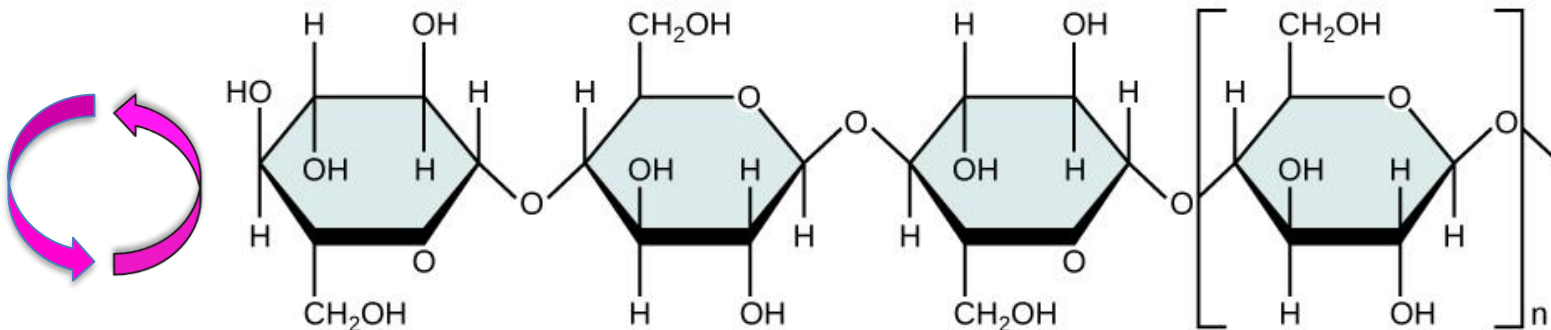
Amylopectin (75-80%) is composed of branched chains of glucose monomers connected by α 1,4 and α 1,6 glycosidic linkages.

Starch is converted into sugars, for example by malting, and fermented to produce ethanol in the manufacture of beer, whisky and biofuel. it is the most common carbohydrate in human diets, and is contained in large amounts in staple foods such as wheat, potatoes, maize (corn), rice, and cassava

Glycogen is the storage form of glucose in humans and other vertebrates and is made up of monomers of glucose. Glycogen is the animal equivalent of starch and is a highly branched molecule usually stored in liver and muscle cells.

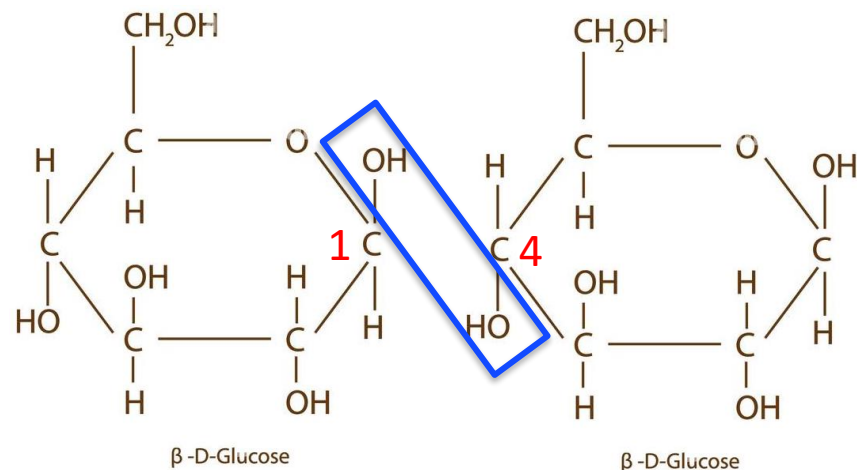
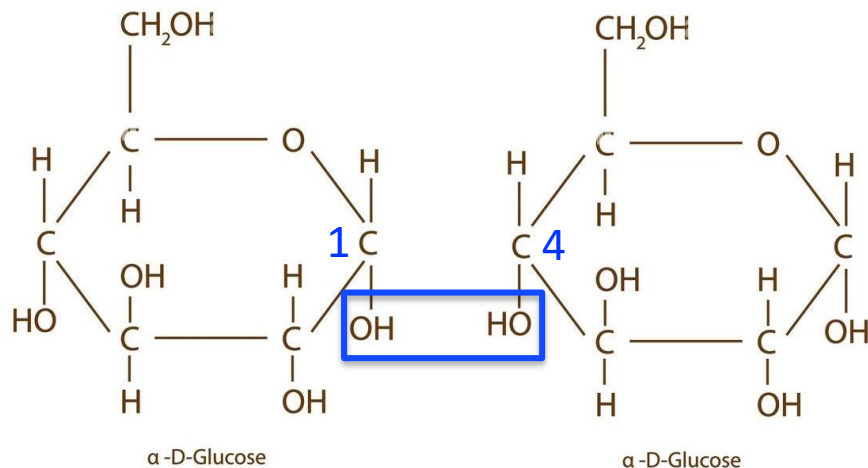
Cellulose

- Cellulose is an important **structural component** of the **primary cell wall** of green plants. Cellulose is the **most abundant** organic polymer on Earth.
- Some species of bacteria secrete it to form **biofilms**.
- Ruminants** and termites, **can digest cellulose** with the help of symbiotic micro-organisms in their guts, such as **Ruminants** and termites, **can digest cellulose** with the help of symbiotic micro-organisms in their guts, such as *Ruminococcus flavefaciens*, *Ruminococcus albus* and *Fibrobacter succinogenes* are the major cellulolytic bacteria and *Trichonympha* a protist.
- In human, cellulose is a **non-digestible constituent** of insoluble **dietary fiber**, acting as a **hydrophilic bulking agent** for feces and potentially aiding in defecation.
- Paper products; cellulose fibers in textile; Microcrystalline cellulose- inactive fillers in drug tablets; building insulation; nitrocellulose (cellulose nitrate).

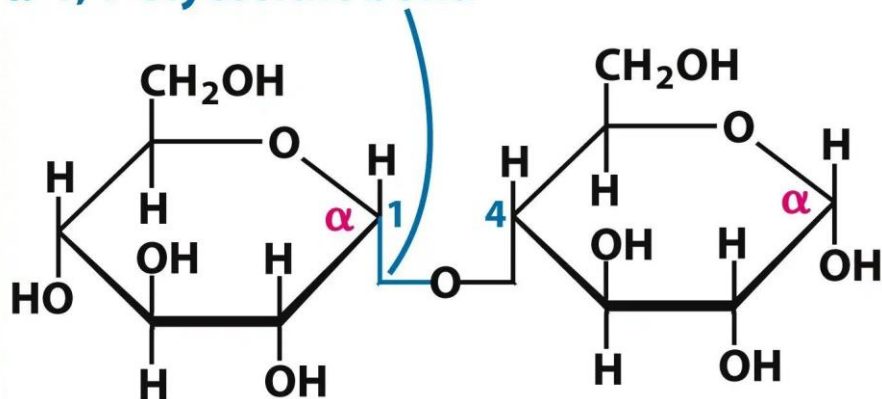


In cellulose, glucose monomers are linked in unbranched chains by **β 1-4 glycosidic linkages**

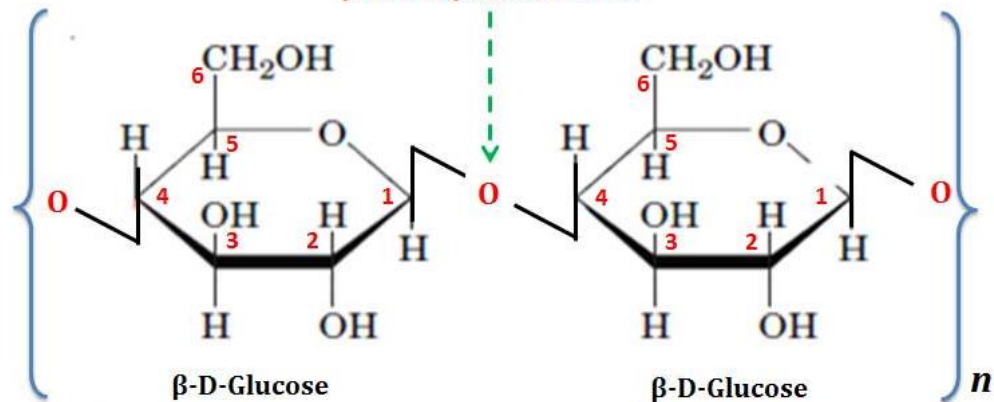
alpha-1,4- and beta-1,4-glycosidic bond



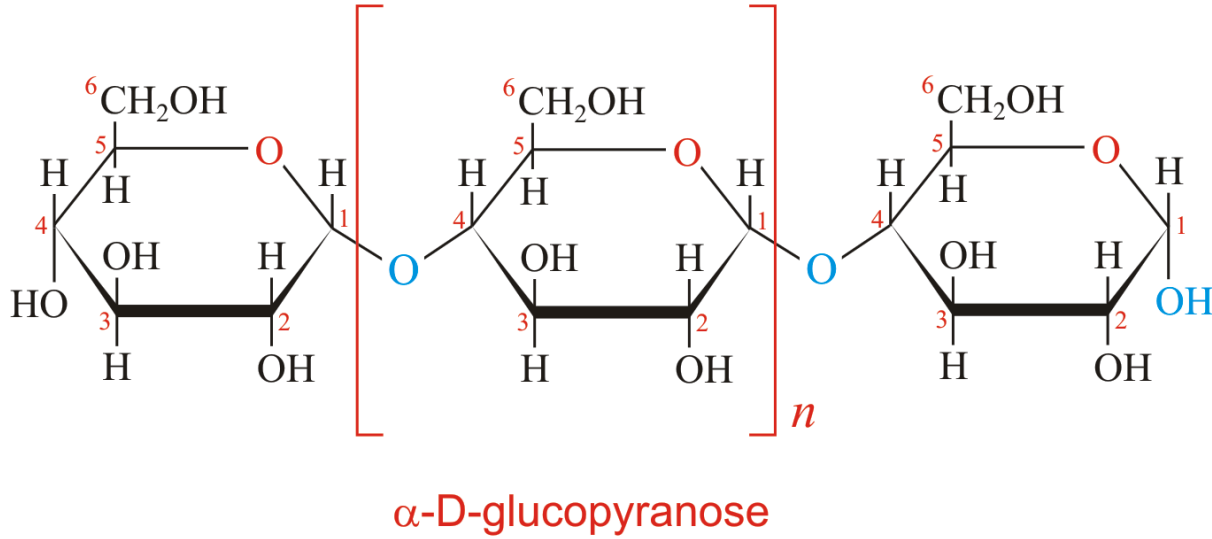
α -1,4-Glycosidic bond



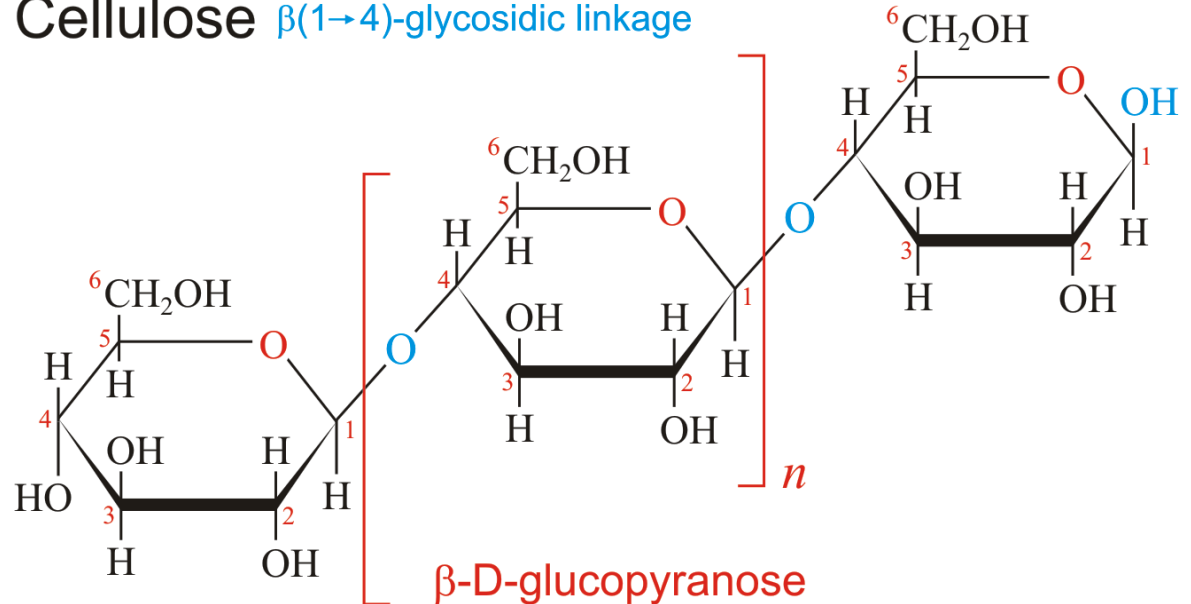
β 1 $\rightarrow$ 4 Glycosidic Bond



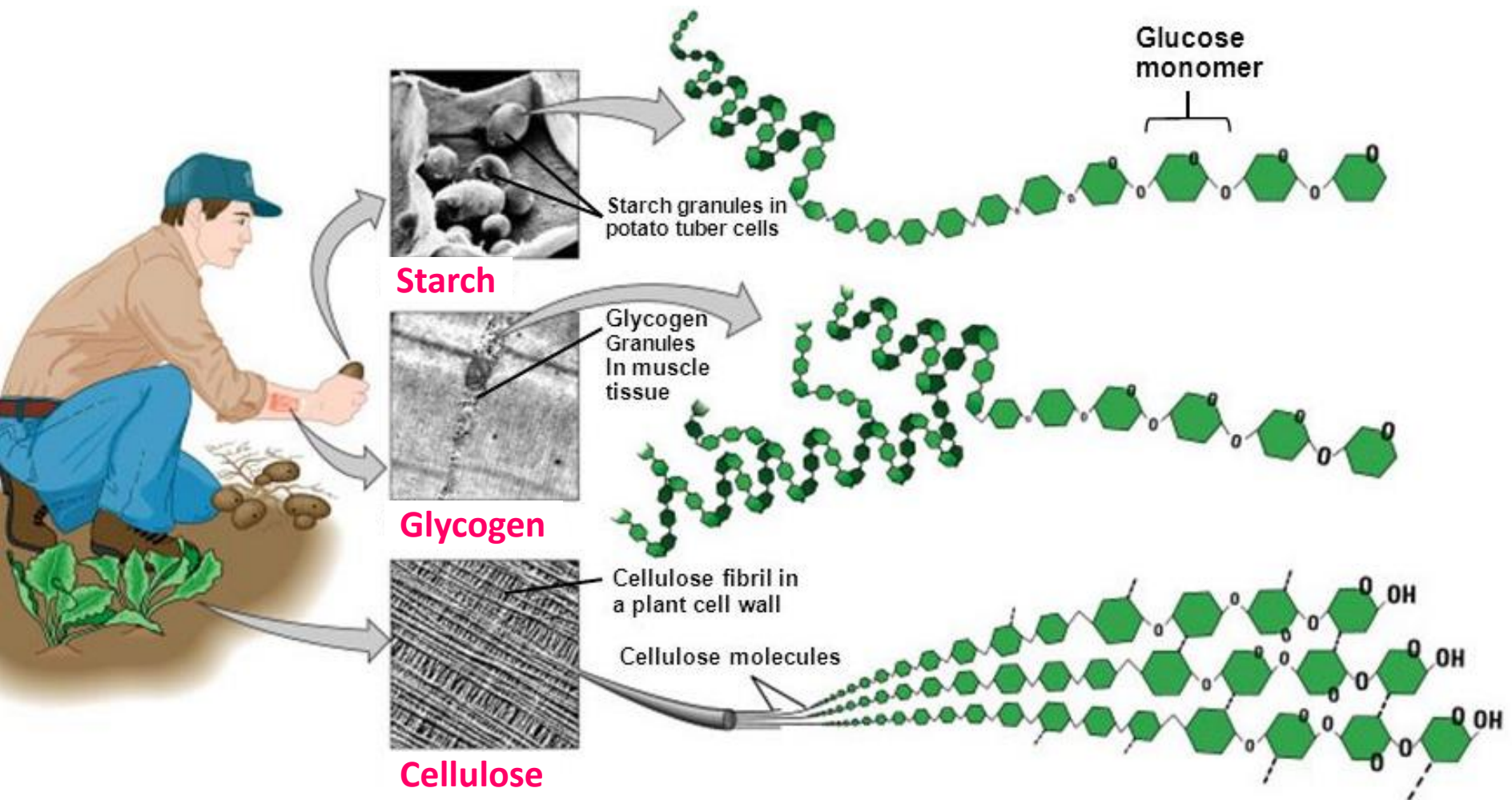
Starch $\alpha(1\rightarrow4)$ -glycosidic linkage

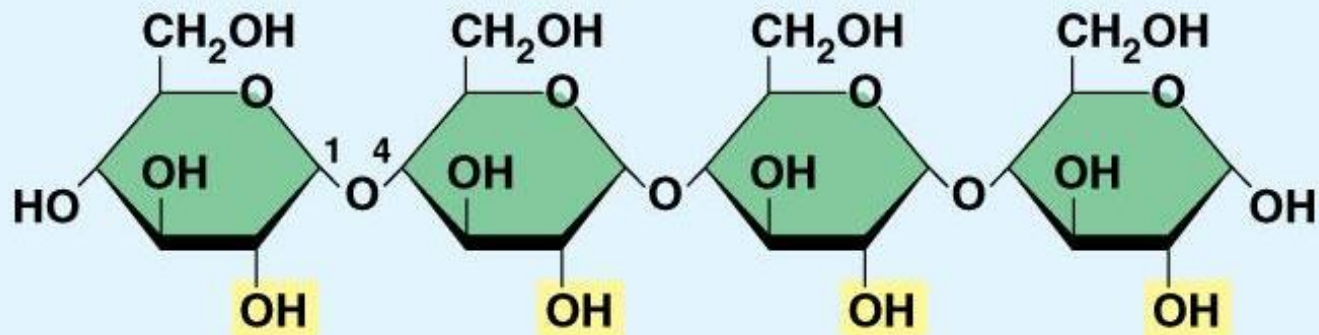


Cellulose $\beta(1\rightarrow4)$ -glycosidic linkage

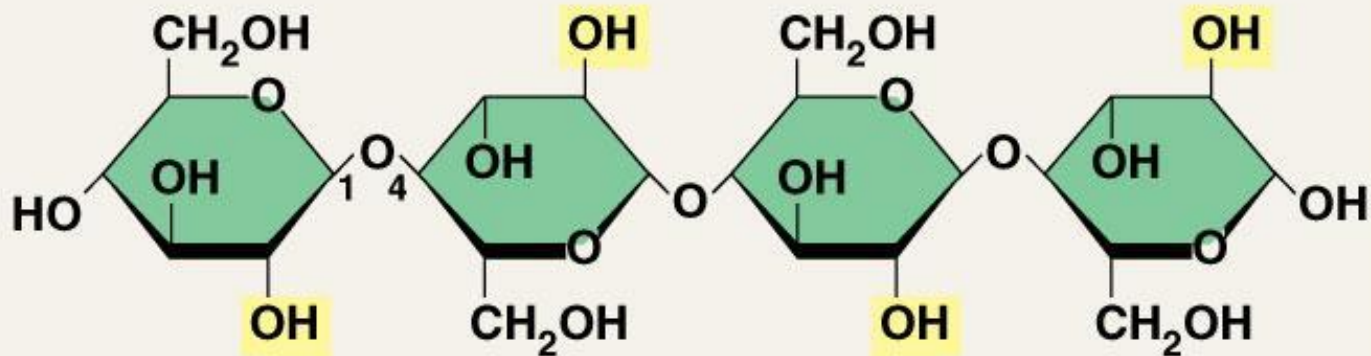


Common Polysaccharides

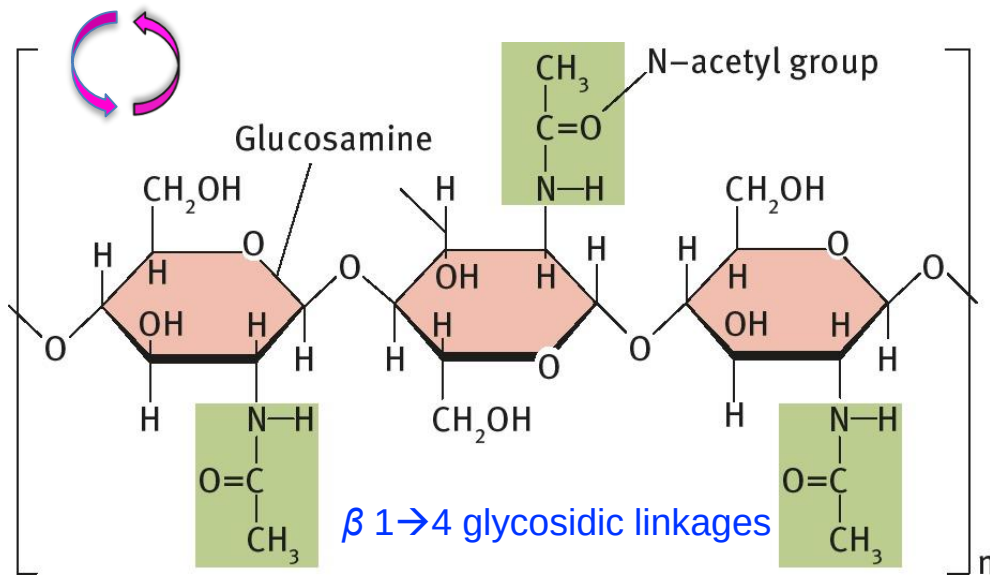




(b) Starch: 1-4 linkage of α glucose monomers



(c) Cellulose: 1-4 linkage of β glucose monomers



Chitin is a long-chain polymer of N-acetylglucosamine, an amide derivative of glucose. Found in cell walls in fungi, the exoskeletons of arthropods such as crustaceans and insects, and gladii of mollusks.

Starch and Glycogen

- Starch is energy storage molecule in plants
- Glycogen is energy storage molecule in animals.
- Starch and glycogen can be digested by animals.

Cellulose

- Different bond formed than starch
- Structural component in plants
- **Cannot be digested** by animals
- *Ruminococcus flavefaciens*, and *Ruminococcus albus* **digest cellulose**

Chitin

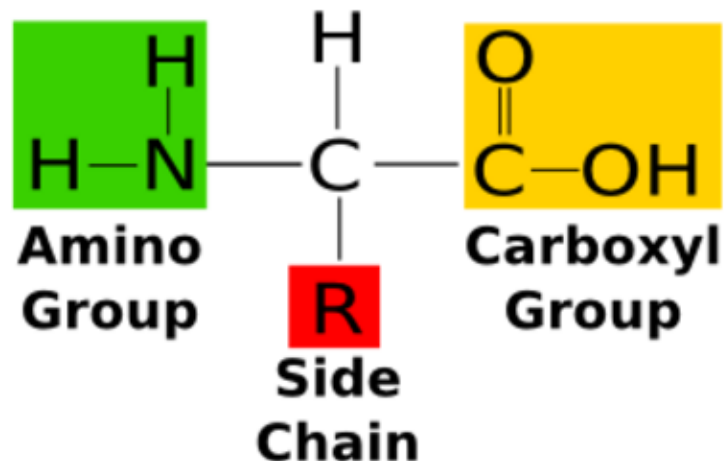
- Partly derived from non-sugars (nitrogen)
- Composes **exoskeletons** of insects



Proteins are polymers of amino acids

Amino Acids: Building Block of Proteins

- Thousands of different kinds of proteins are made from only **twenty** monomers, called **amino acids**.
- Contains an amine group (NH_3) (basic/positive)
- A carboxyl group (COOH) (acidic/negative)
- One or more atoms called an “R group”
- All three groups are attached to the same carbon atom



Amino acid

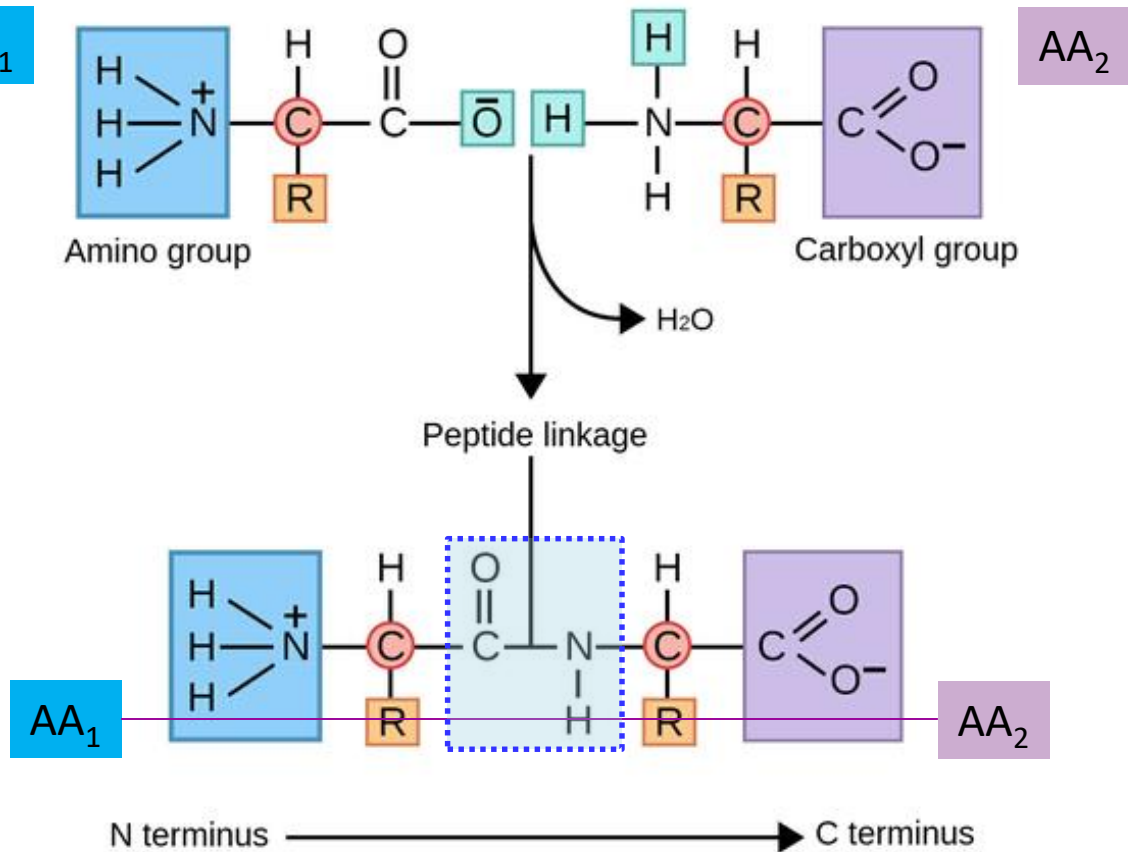
- Amino acids contain **carbon (C)**, **hydrogen (H)**, **oxygen (O)**, **nitrogen (N)** and **sulfur (S)**
- There are 20 different kinds of amino acids (AA)
- Amino acids are divided into two groups-

Essential: must be supplied in the diet.

Non-essential: not supplied in the diet.

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Asparagine	Asparatate
Leucine	Glutamine	Cysteine
Methionine	Glycine	Glutamate
Phenylalanine	Proline	
Threonine	Serine	
Tryptophan	Tyrosine	
Valine		
Lysine		

Peptide bond is an amide type of covalent chemical bond linking two consecutive alpha-amino acids



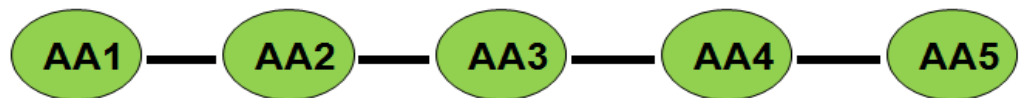
$AA + AA \rightarrow$ Dipeptide



$AA + \text{Dipeptide} \rightarrow$ Tripeptide



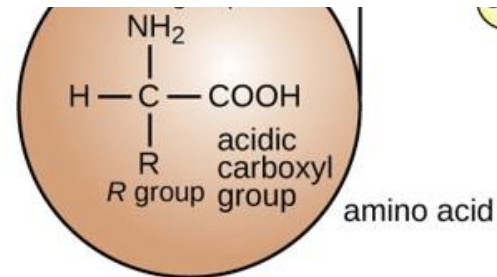
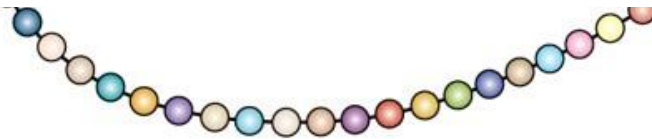
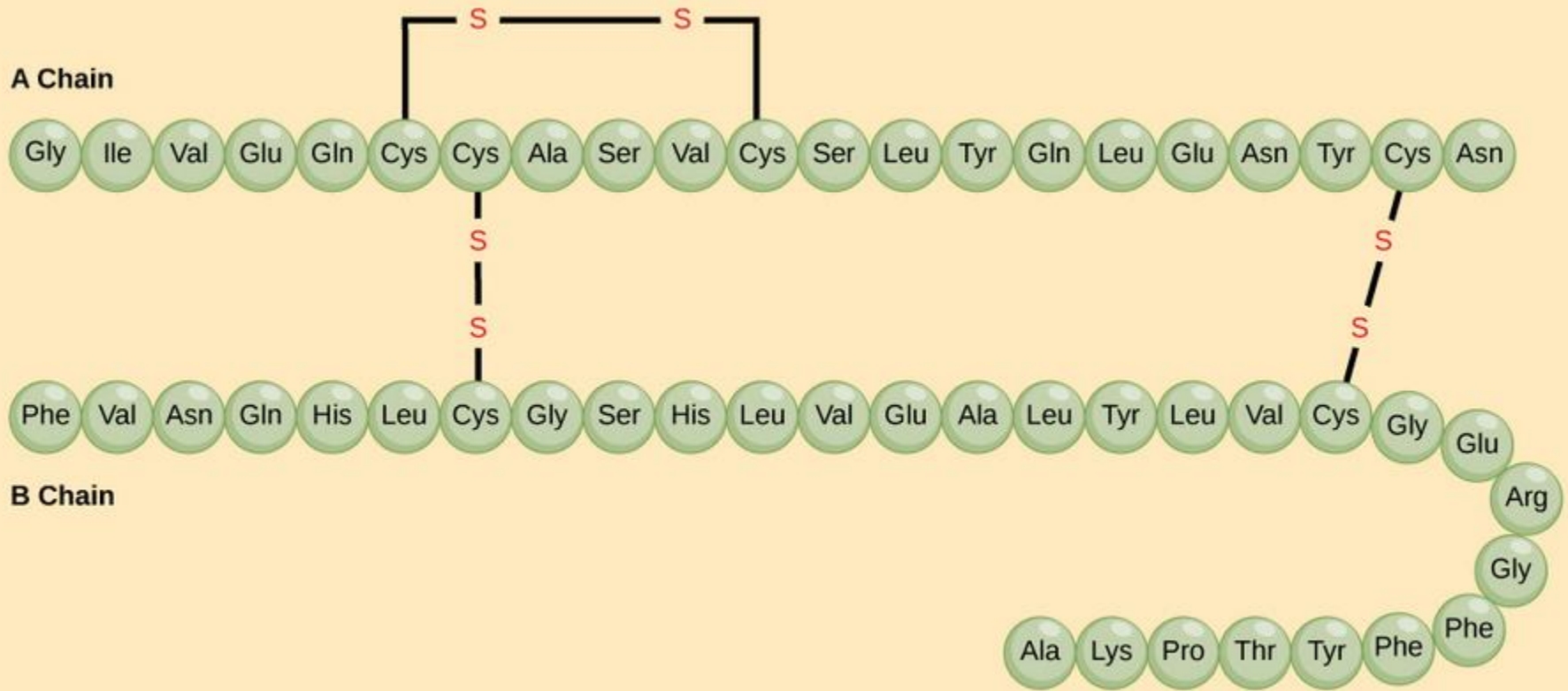
$AA + AA + \dots + \text{Tripeptide} \rightarrow$ Polypeptide



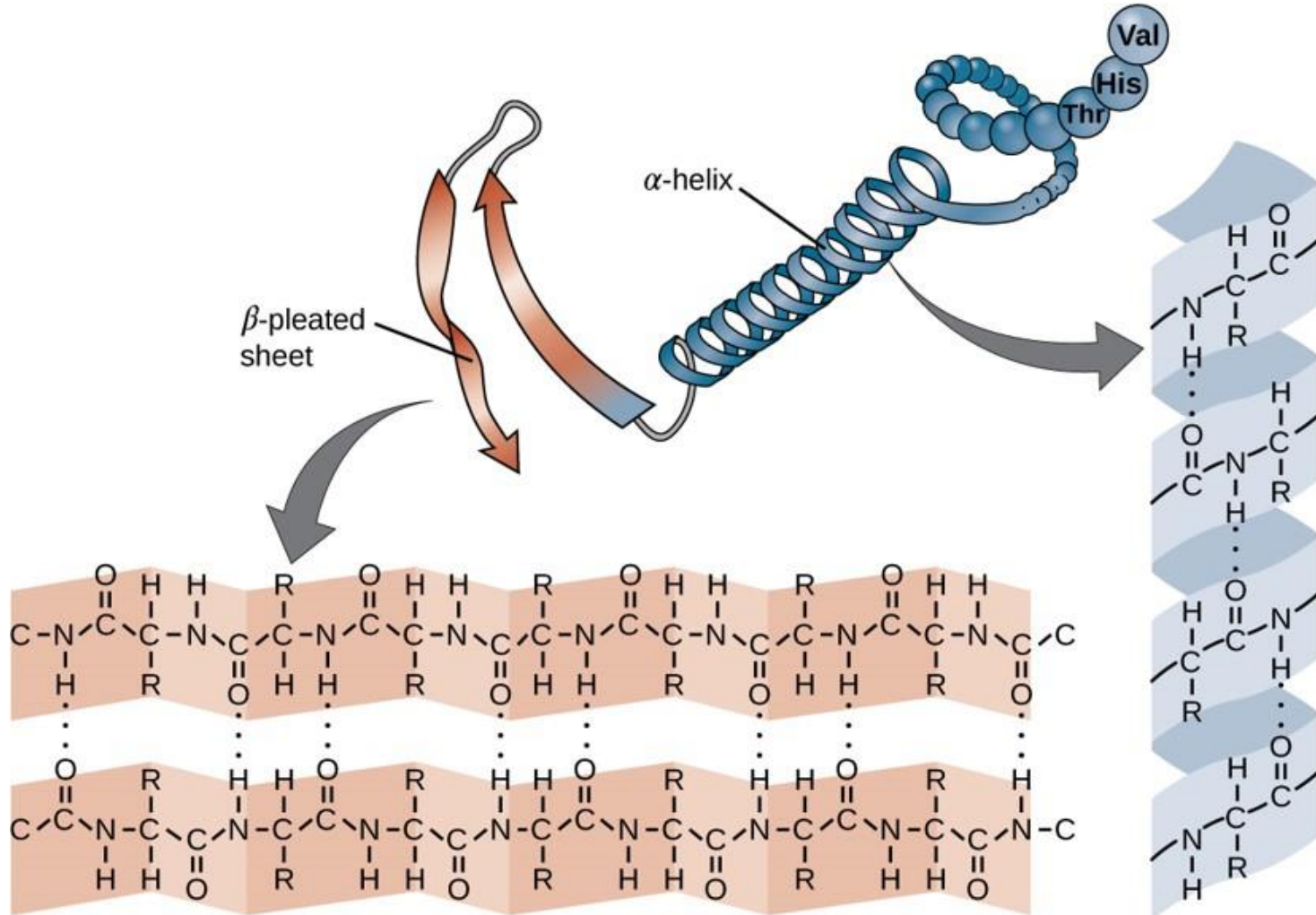
Levels of protein structure

1. **Primary structure:** The linear arrangement of amino acids in a protein
2. **Secondary structure:** Areas of folding or coiling within a protein. e.g. α -helices and β -pleated sheets
3. **Tertiary structure:** Final three-dimensional structure of a protein, which results from a large number of non-covalent interactions between amino acids.
4. **Quaternary structure:** Non-covalent interactions that bind multiple polypeptides into a single, larger protein. e.g. Hemoglobin

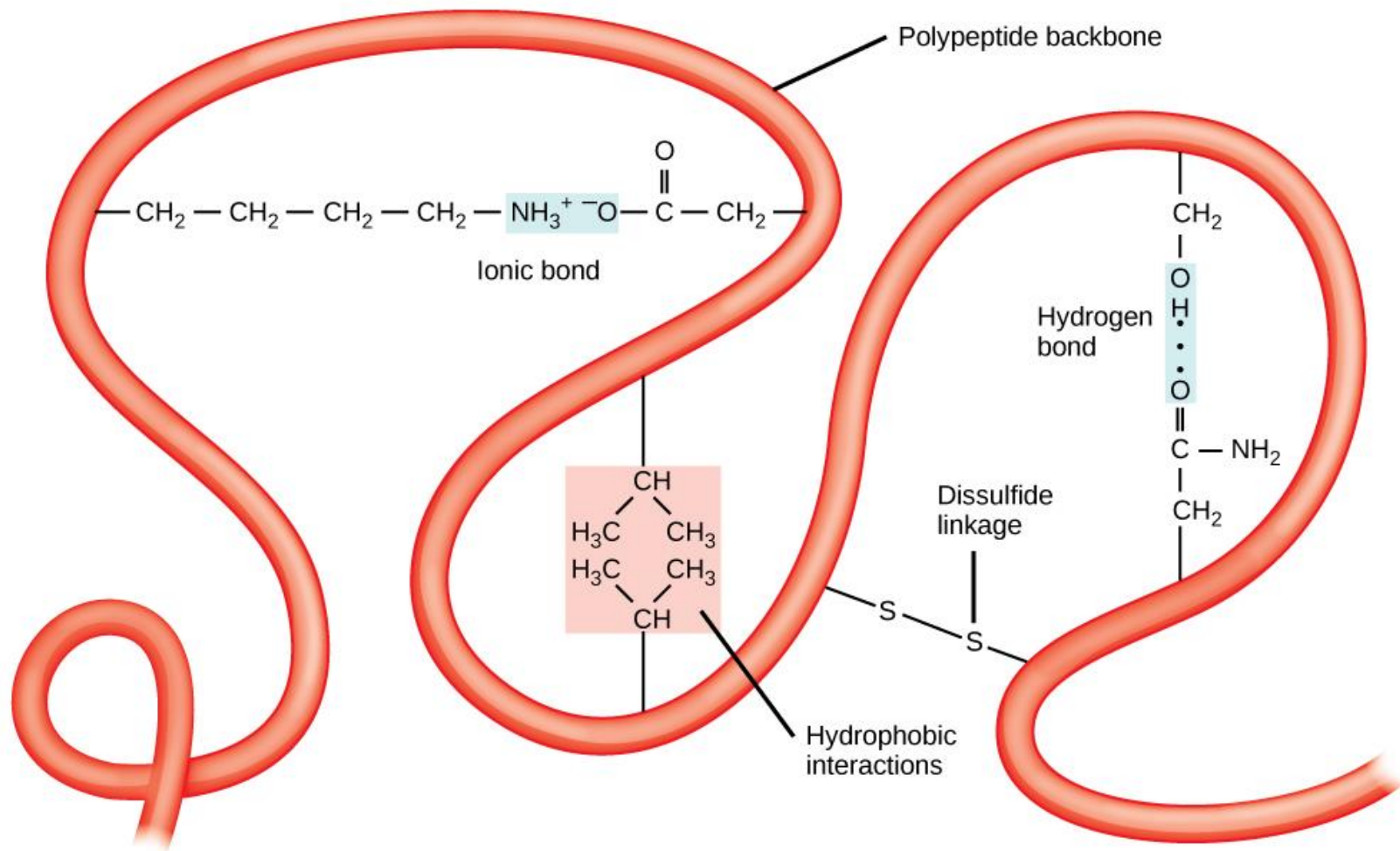
Primary Structural Features



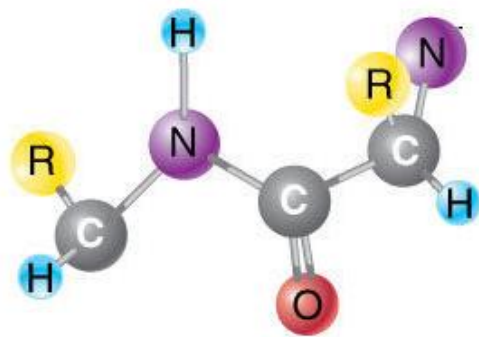
Secondary Structural Features



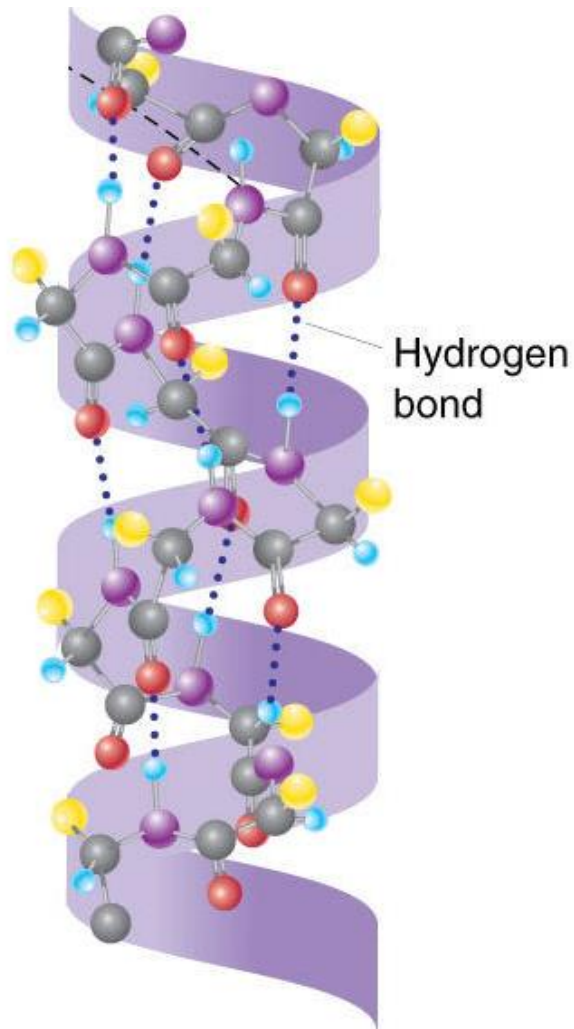
Tertiary Protein Structure



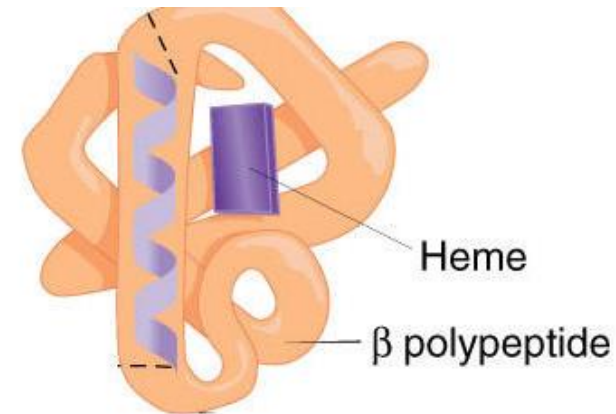
Levels of Protein Structure



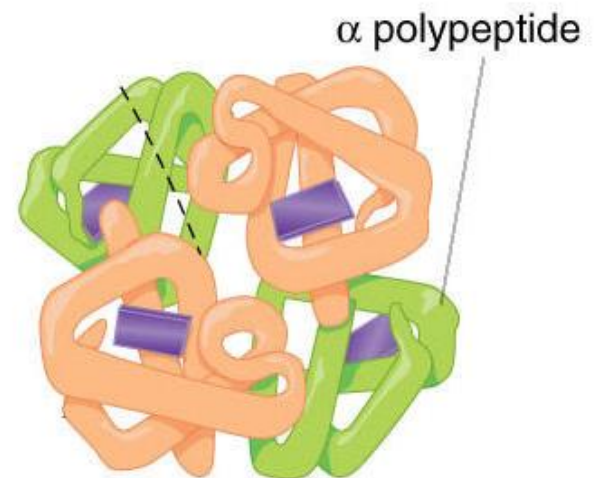
(a) Primary structure



(b) Secondary structure

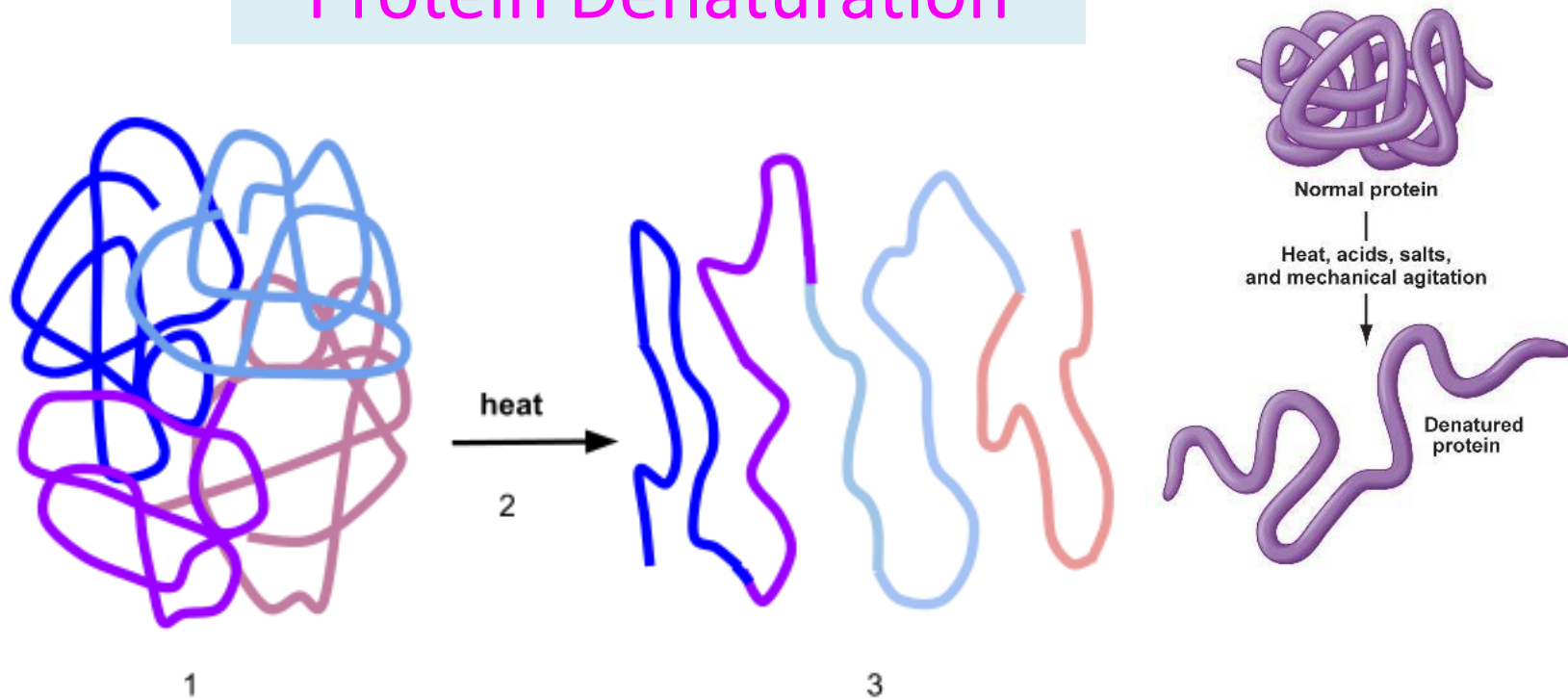


(c) Tertiary structure



(d) Quaternary structure—

Protein Denaturation



- (1) the correctly folded intact protein.
- (2) applies heat to the system that is above the threshold of maintaining the intramolecular protein interactions.
- (3) the unfolded or denatured protein.

Primary structure is unchanged by denaturing.

Main functions of proteins

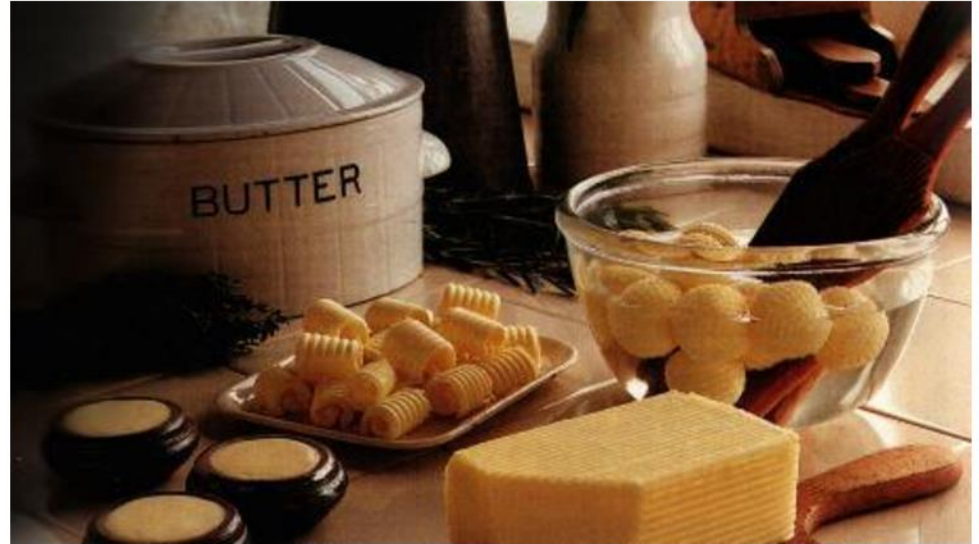
- Protein's main function is to build, maintain and repair all our body tissues
- Protein can also be used as energy source by body

Biological functions of proteins

- Protein acts as **storage material** of food and energy.
- Many proteins are **enzymes** that catalyze biochemical reactions
- Proteins are **molecular instrument** through which genetic information is expressed.
- They act as **antibodies** to prevent disease.
- The **milk proteins** help the growth of infant mammals.

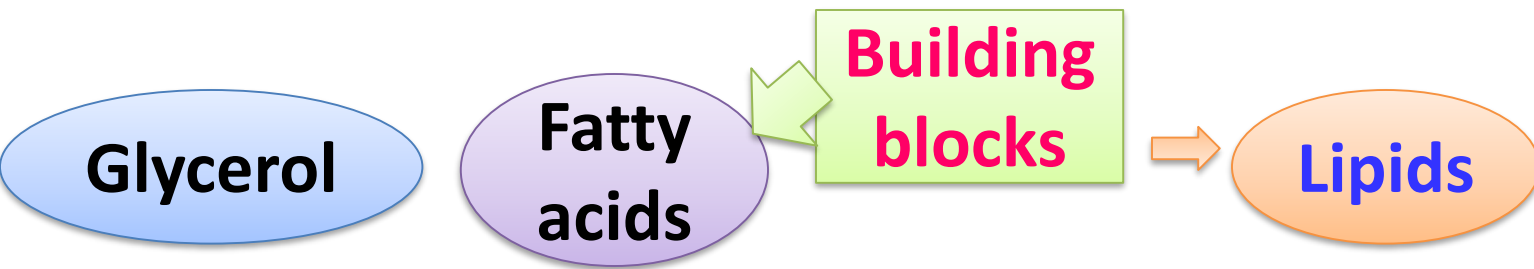
For detail functions see the notes underneath...

Lipids: Fats & Oils



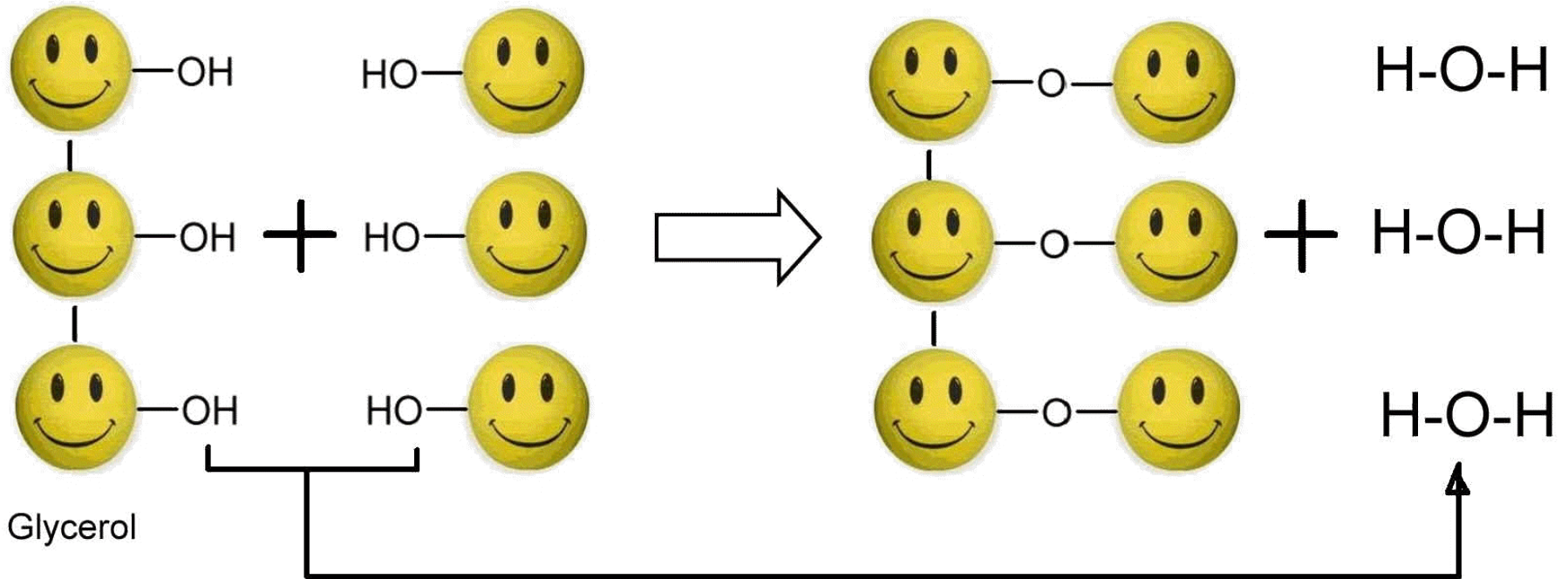
Lipids

- Composed of Carbon, Hydrogen, and Oxygen
- Differ from carbohydrates: no specific ratio (C:H:O)
- Building blocks are fatty acids and glycerol.
- Energy storage molecules “store the most energy” 2X of Carbohydrates. 1g of Carbohydrate or Protein = 4 cal, 1g of Fat = 9 cal
- Structurally heterogeneous, Not soluble in water
- Are soluble in hydrophobic solvents.
- Examples: 1. Fats 2. Oils 3. Phospholipids
4. Waxes 5. Steroid hormones 6. Triglycerides



3 Fatty Acid Chains

A complete lipid



3 Water molecules are removed
forming 3 oxygen bridges

Dehydration Synthesis of a Lipid



CAN BE



GOOD

Unsaturated Fats

VS



BAD

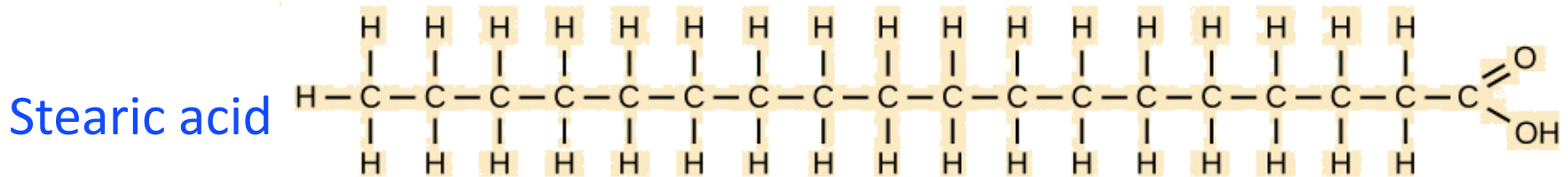
Saturated Fats



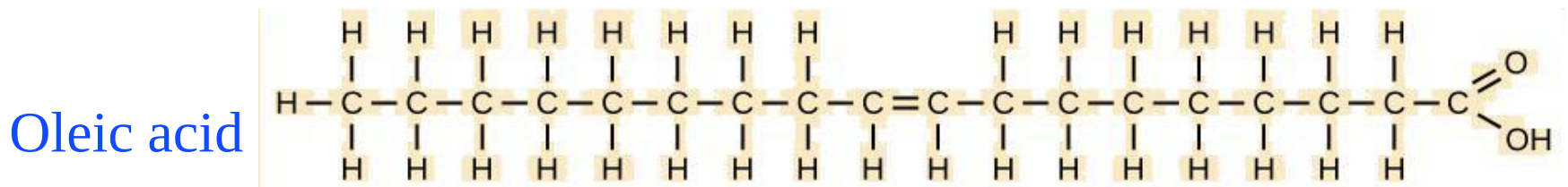
Fatty acid

- A fatty acid is a carboxylic acid with a long aliphatic chain, which is either saturated or unsaturated. Most naturally occurring fatty acids.

Saturated Fatty acid: In a fatty acid chain, if there are only single bonds between neighbouring carbons in the hydrocarbon chain.

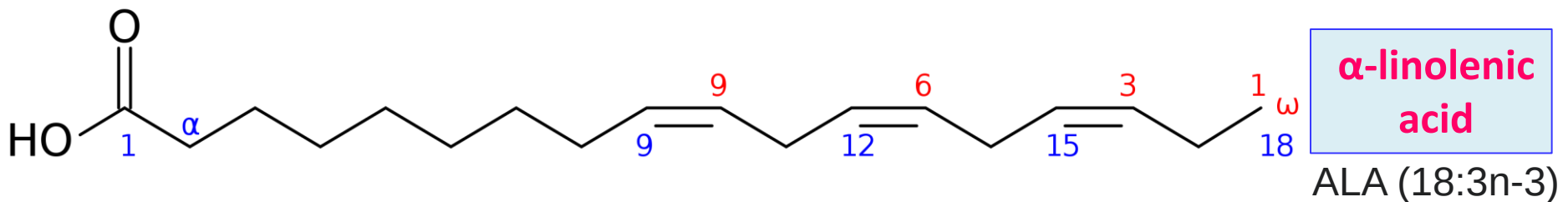
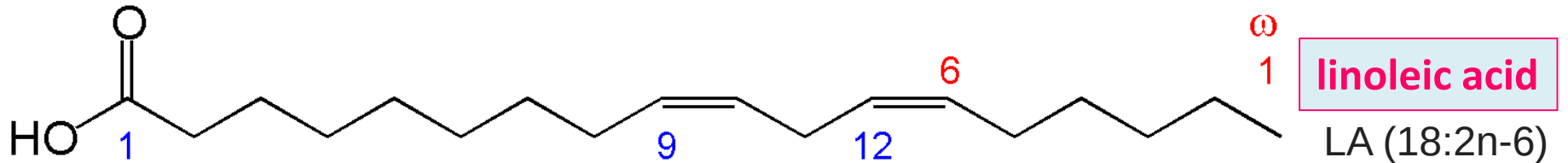


Unsaturated Fatty acid: When the hydrocarbon chain contains a double bond, between neighbouring carbons.

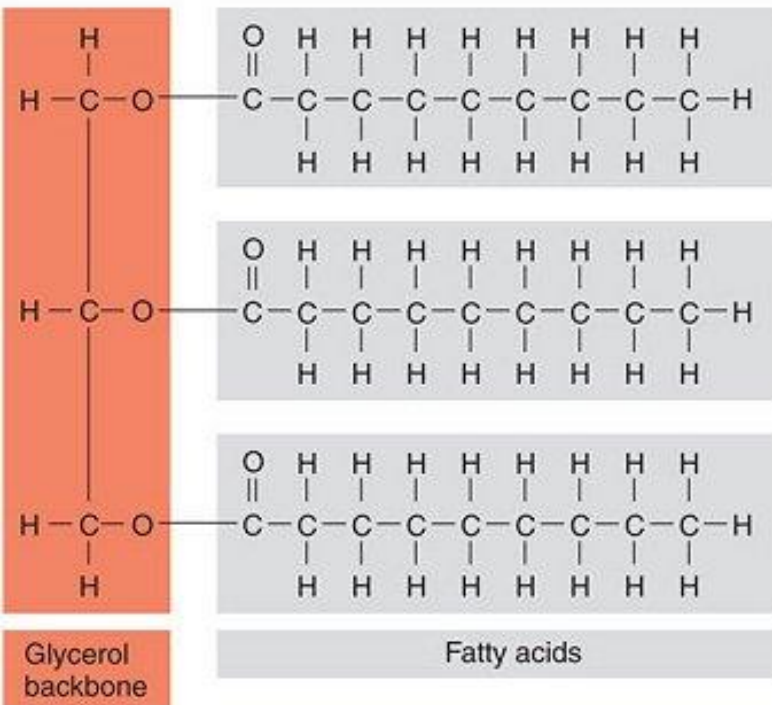


Type of Fatty Acid	Double Bonds	Diagram
Saturated	None	
Monounsaturated	One	
Polyunsaturated	Multiple (>1)	

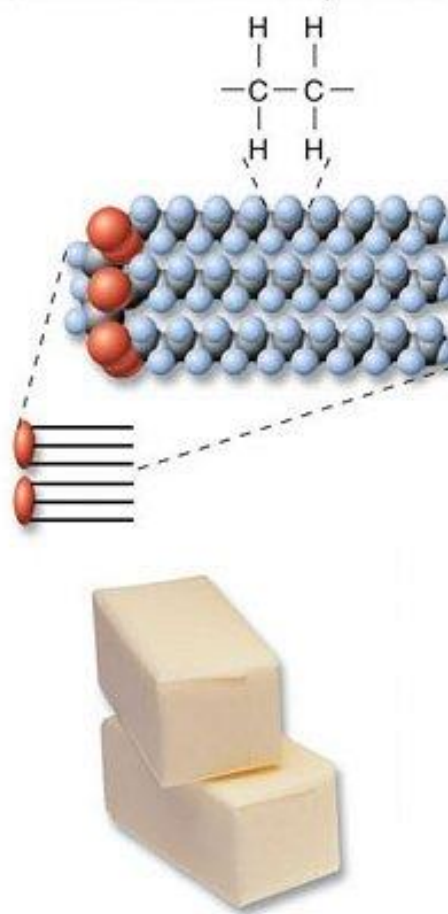
The term **essential fatty acids (EFA)** refers to **those polyunsaturated fatty acids (PUFA) that must be provided by foods because these cannot be synthesized in the body yet are necessary for health**. Only two fatty acids are known to be essential for humans: alpha-linolenic acid (18:3n-3; an omega-3 fatty acid) and linoleic acid (18:2n-6; an omega-6 fatty acid).



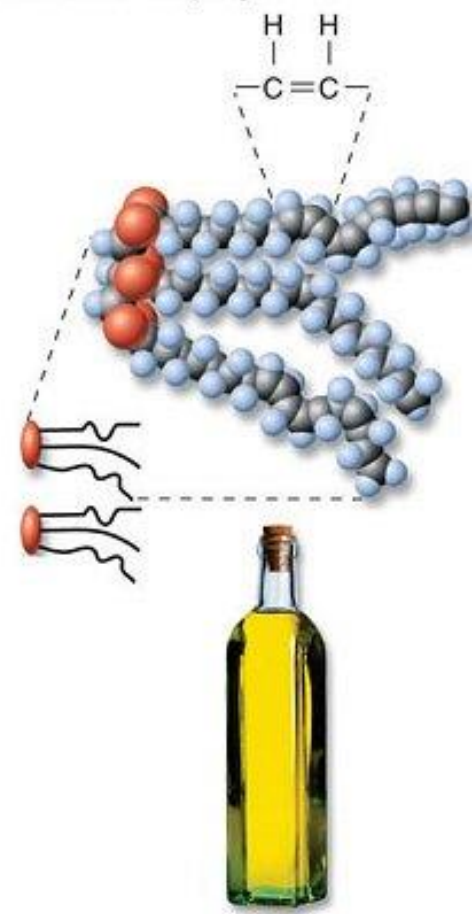
Oils and fat



(a) Fat molecule (triacylglycerol)



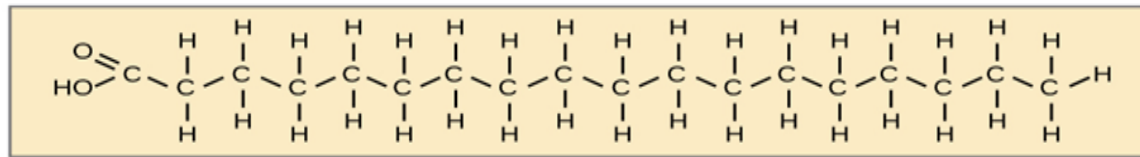
(b) Hard fat (saturated): Fatty acids with single bonds between all carbon pairs



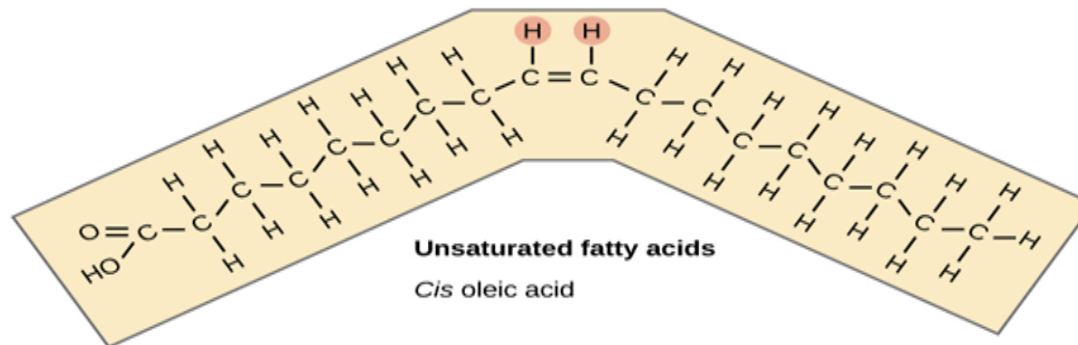
(c) Oil (unsaturated): Fatty acids that contain double bonds between one or more pairs of carbon atoms

Fat and oils

- Most unsaturated fats are liquid at room temperature and are called oils. If there is one double bond in the molecule, then it is known as a **monounsaturated fat** (e.g., olive oil), and if there is more than one double bond, then it is known as a **polyunsaturated fat** (e.g., canola oil).
- *Cis and trans* indicate the configuration of the molecule around the double bond. If hydrogens are present in the same plane, it is referred to as a **cis fat**; if the hydrogen atoms are on two different planes, it is referred to as a **trans fat**.
- The *cis double* bond causes a bend or a “kink” that **prevents the fatty acids from packing tightly**, keeping them liquid at room temperature. Olive oil, corn oil, canola oil, and cod liver oil are examples of unsaturated fats.
- Unsaturated fats help to lower blood cholesterol levels where-as saturated fats contribute to plaque formation in the arteries.

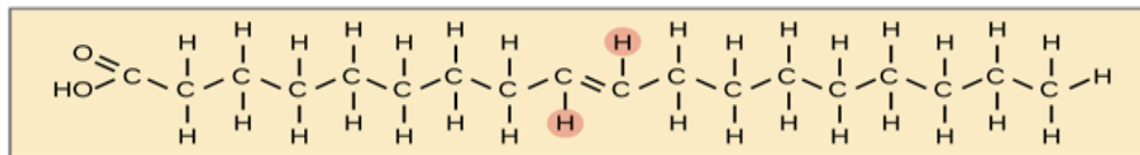


Saturated fatty acid
Stearic acid



Unsaturated fatty acids
Cis oleic acid

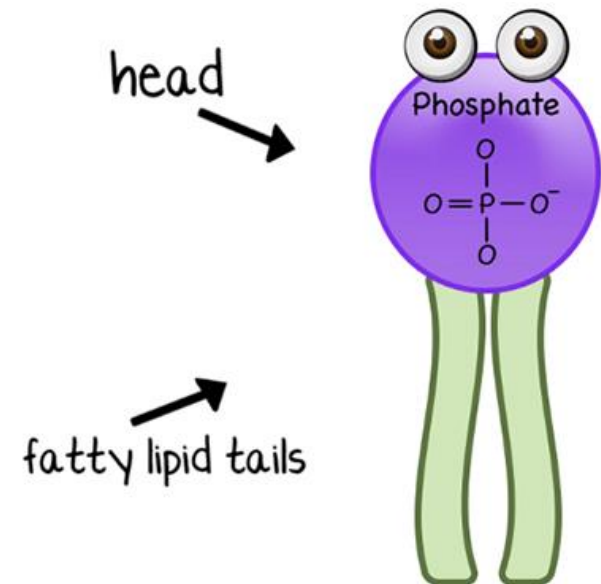
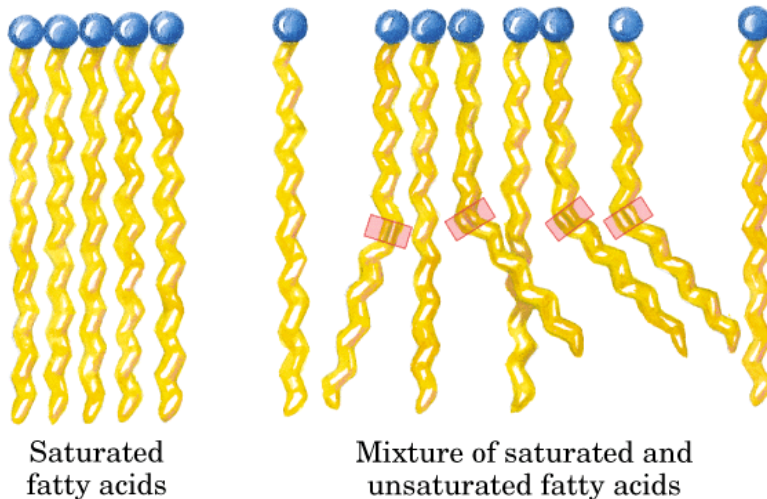
Trans oleic acid



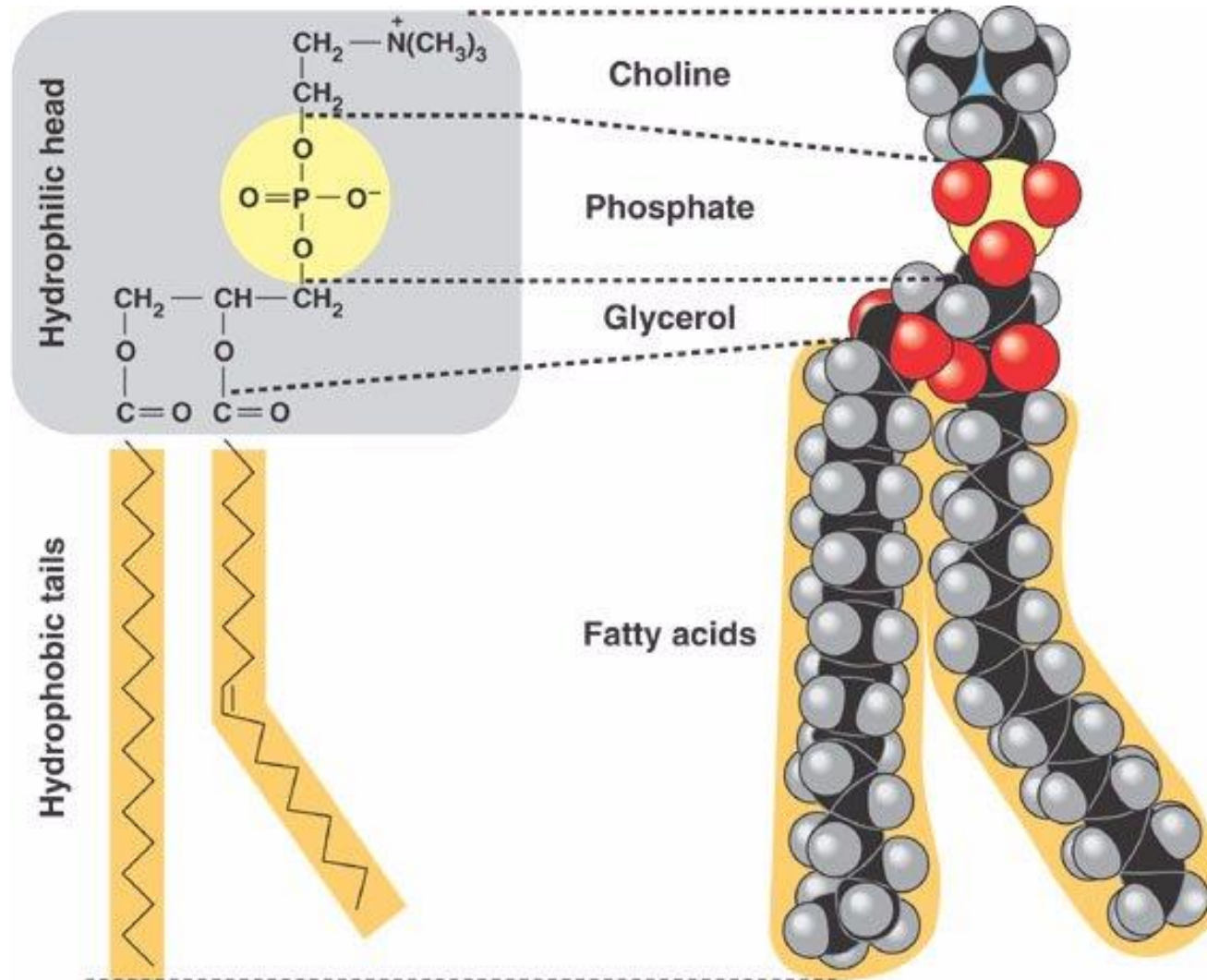
Unsaturated fatty acids have one or more double bonds. Each double bond may be in a *cis or trans* configuration. A cis double bond causes a kink in the chain.

Phospholipids

- ❑ Phospholipids are a variation on the triacylglycerol theme in which
 - One fatty acid is replaced with a phosphate group, which in turn is bound to additional functional groups.
- ❑ Structurally and functionally, the important thing about phospholipids is that
 - These molecules are simultaneously hydrophobic (at one end, the fatty acid end) and hydrophilic (at the other end, the phosphate end). **Amphiphile**

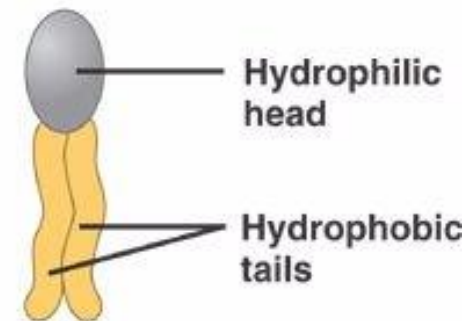


Phospholipids



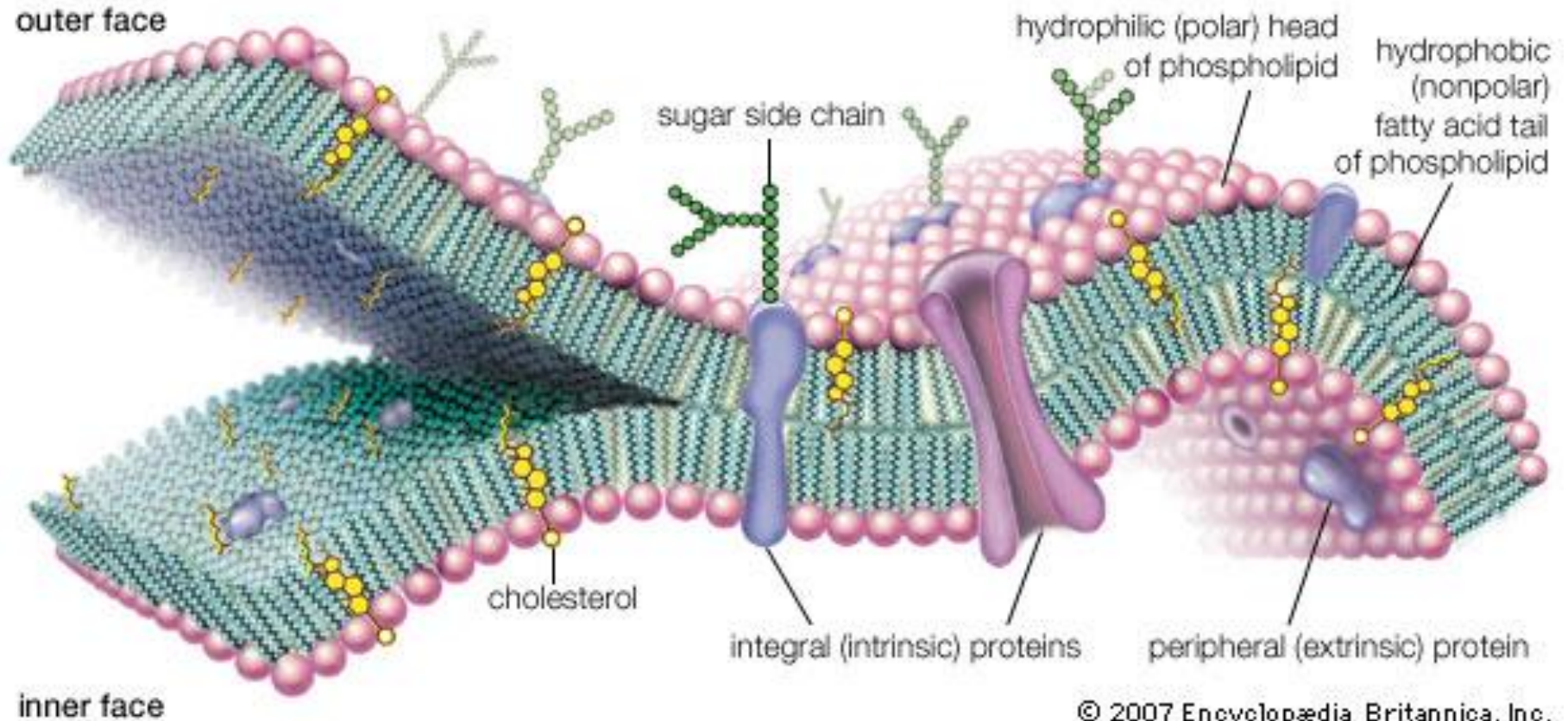
(a) Structural formula

(b) Space-filling model



(c) Phospholipid symbol

- Phospholipids are the most abundant lipids in cell membranes, which have two layers of lipids

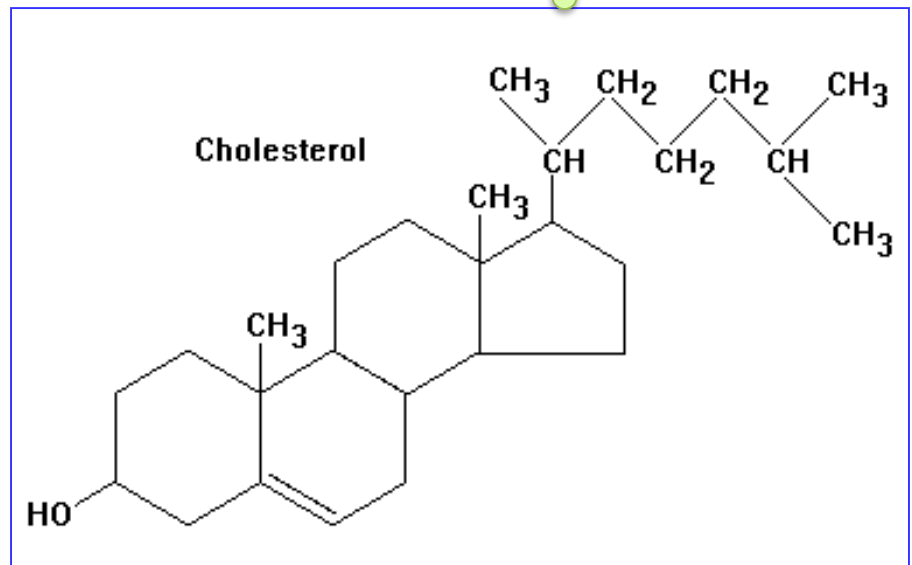


Steroids

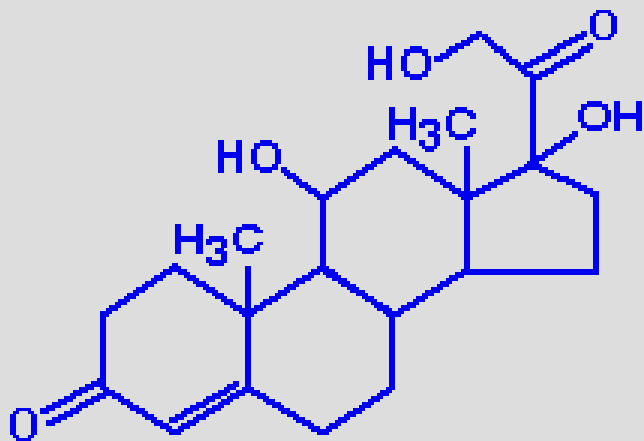
- All steroids possess a common ring structure.
- These ring structures vary by attached functional groups.
- Cholesterol is example of a steroid; cholesterol is a membrane component
- The common steroid structure is the basis of sterol hormones including the human sex hormones (the estrogens and the androgens, including testosterone).

Trans fat ??

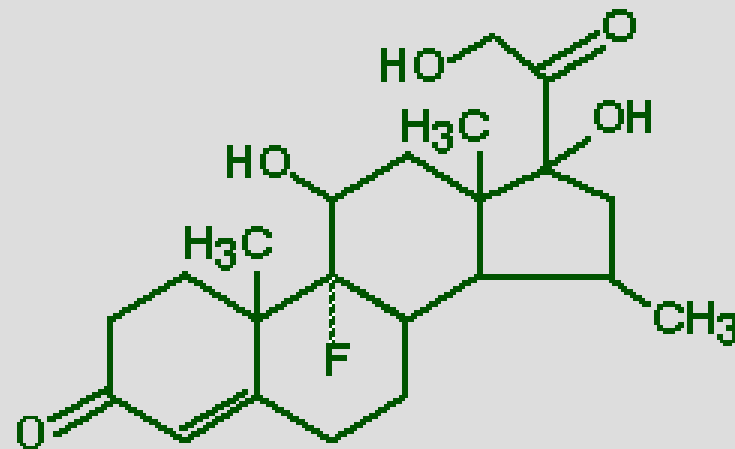
Membrane component



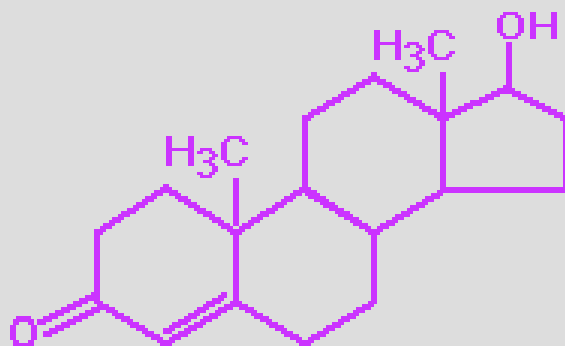
Some Steroid Hormones



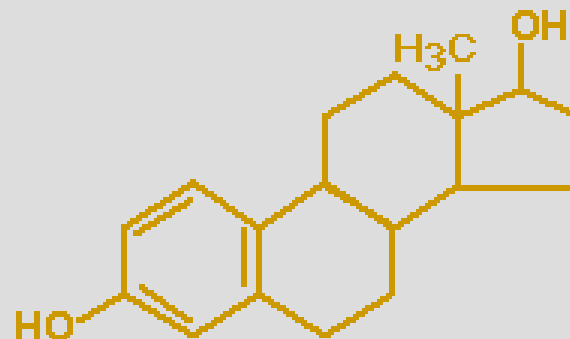
cortisol
(a glucocorticoid)



dexamethasone
(a cortisone analogue)



testosterone
(an androgen)



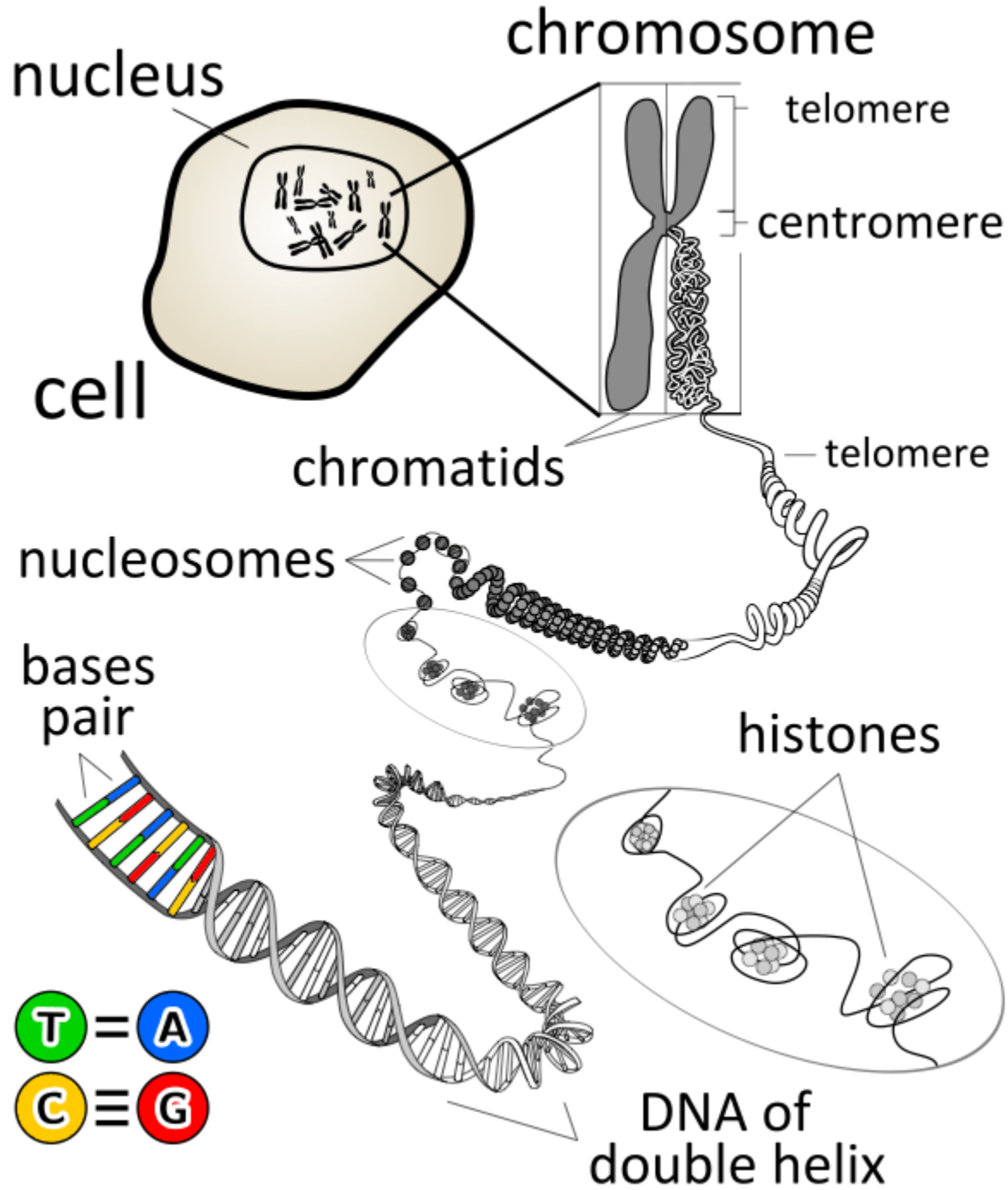
estradiol
(an estrogen)

**Sex specific
hormones**

The fat guidelines

- Limit total fat intake to less than 25–35% of your total calories each day;
- Limit saturated fat intake to <7% of total daily calories;
- Limit trans fat intake to <1% of total daily calories;
- The remaining fat should come from sources of monounsaturated and polyunsaturated fats such as nuts, seeds, fish and vegetable oils; and
- Limit cholesterol intake to less than 300 mg per day, for most people.





Nucleic Acids

- The chemical link between generations
- The source of **genetic information in chromosomes**
- Dictate **amino-acid sequence in proteins**
- Simple units called **nucleotides**, connected in long chains
- **Nucleic acids** are composed of long chains of **nucleotides** linked by **dehydration synthesis**

Nucleic Acids

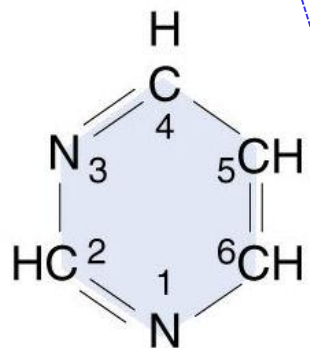
Two types:

- a. Deoxyribonucleic acid (DNA-double helix)
- b. Ribonucleic acid (RNA-single strand)

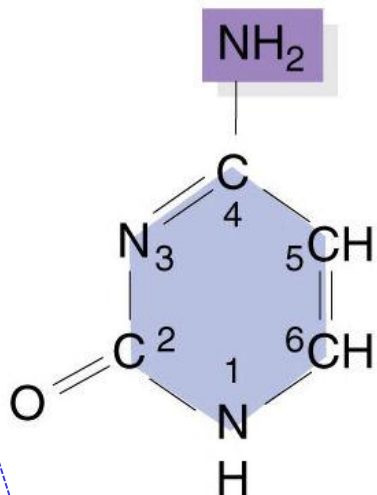
Nucleotides have 3 parts:

- 1. phosphate group (P)
- 2. pentose sugar (5-carbon)
- 3. nitrogenous bases:

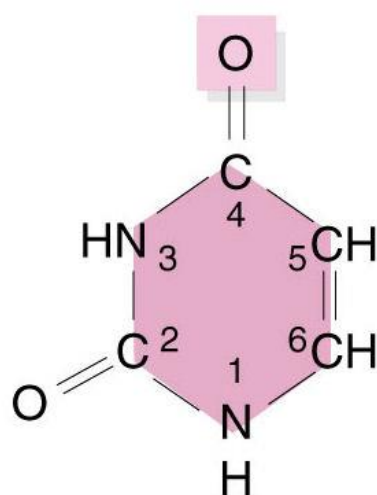
- adenine (A)
- thymine (T) DNA only
- uracil (U) RNA only
- cytosine (C)
- guanine (G)



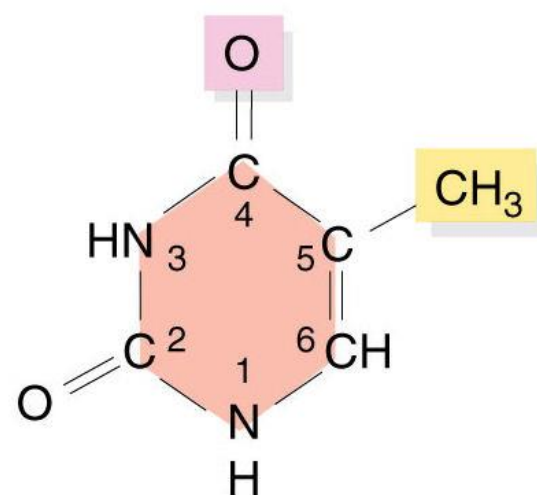
Pyrimidine



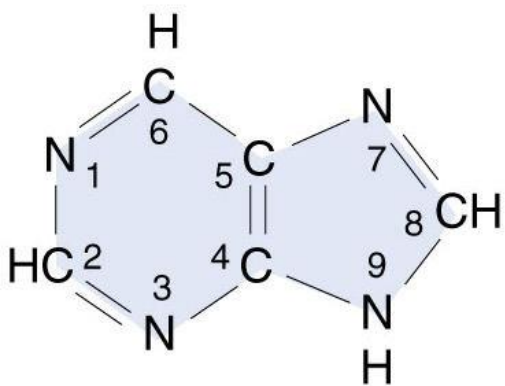
Cytosine (C)



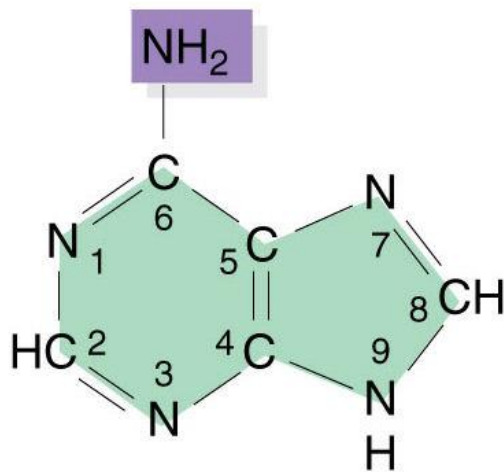
Uracil (U)
(found in RNA)



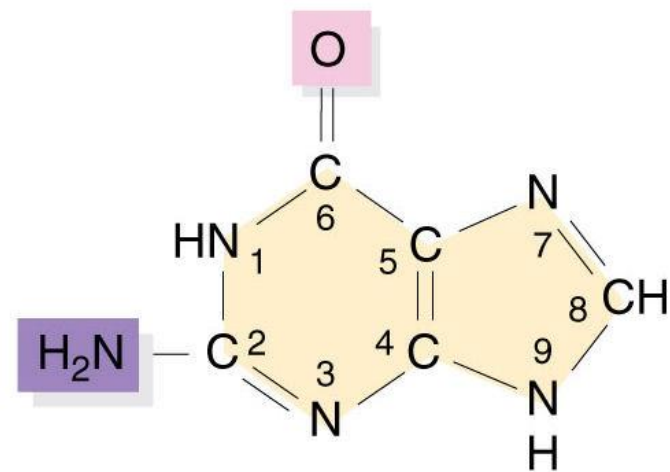
Thymine (T)
(found in DNA)



Purine

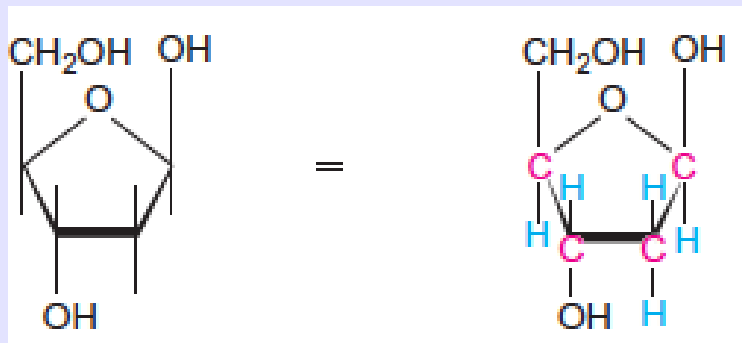
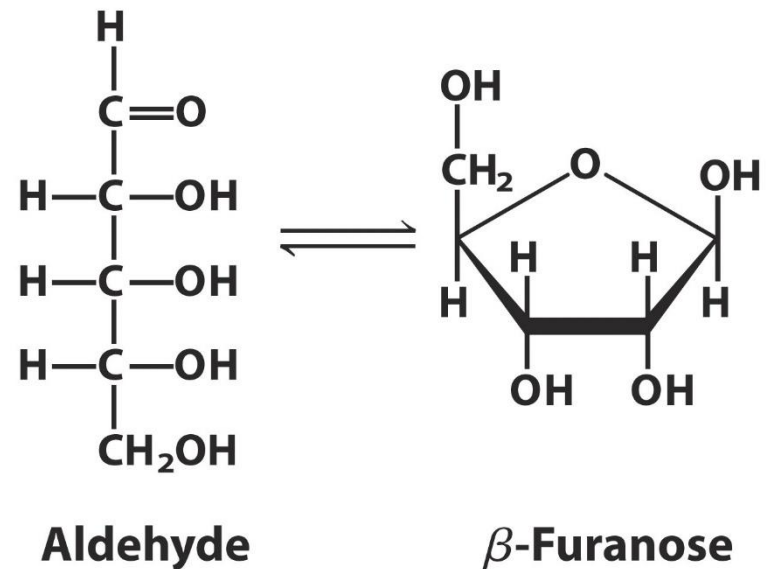
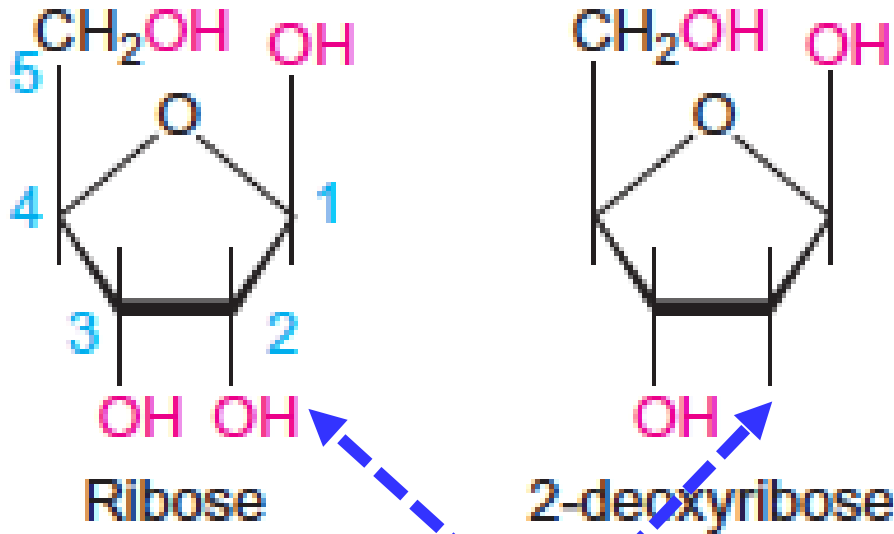


Adenine (A)



Guanine (G)

The sugars of nucleic acids



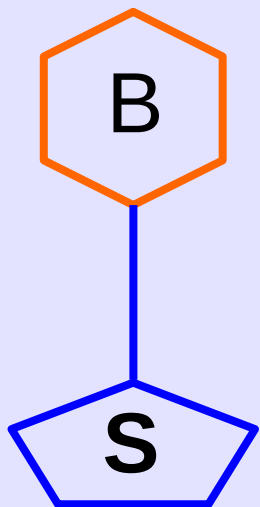
- Note the OH in the 2-position of ribose and its absence in deoxyribose.

Nucleosides (purine)

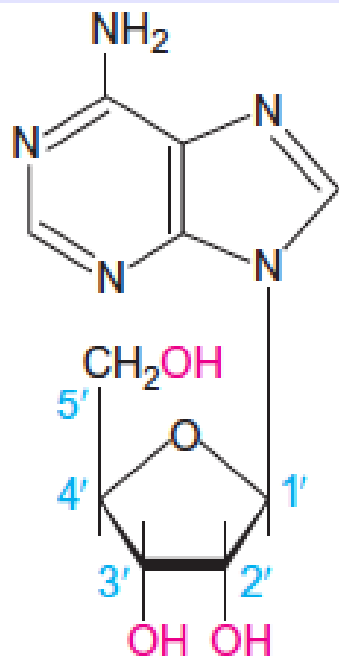
Nucleosides are **glycosylamines** consisting of-

- nucleobase
- ribose or deoxyribose sugar

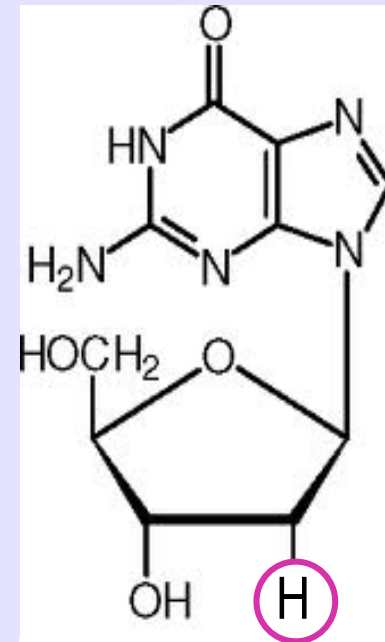
These are bound together via a **beta-glycosidic linkage**



Nucleoside

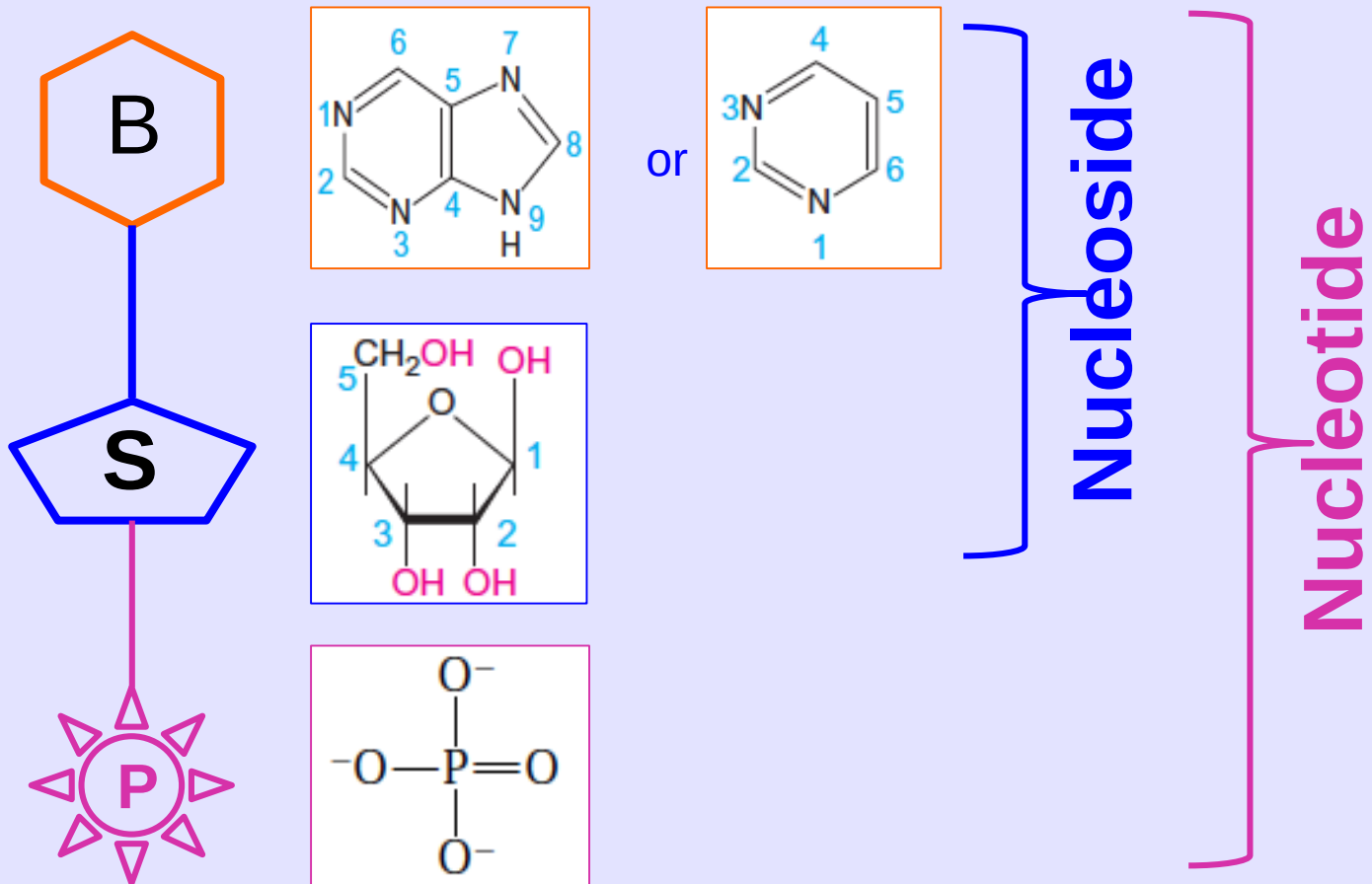


Adenosine

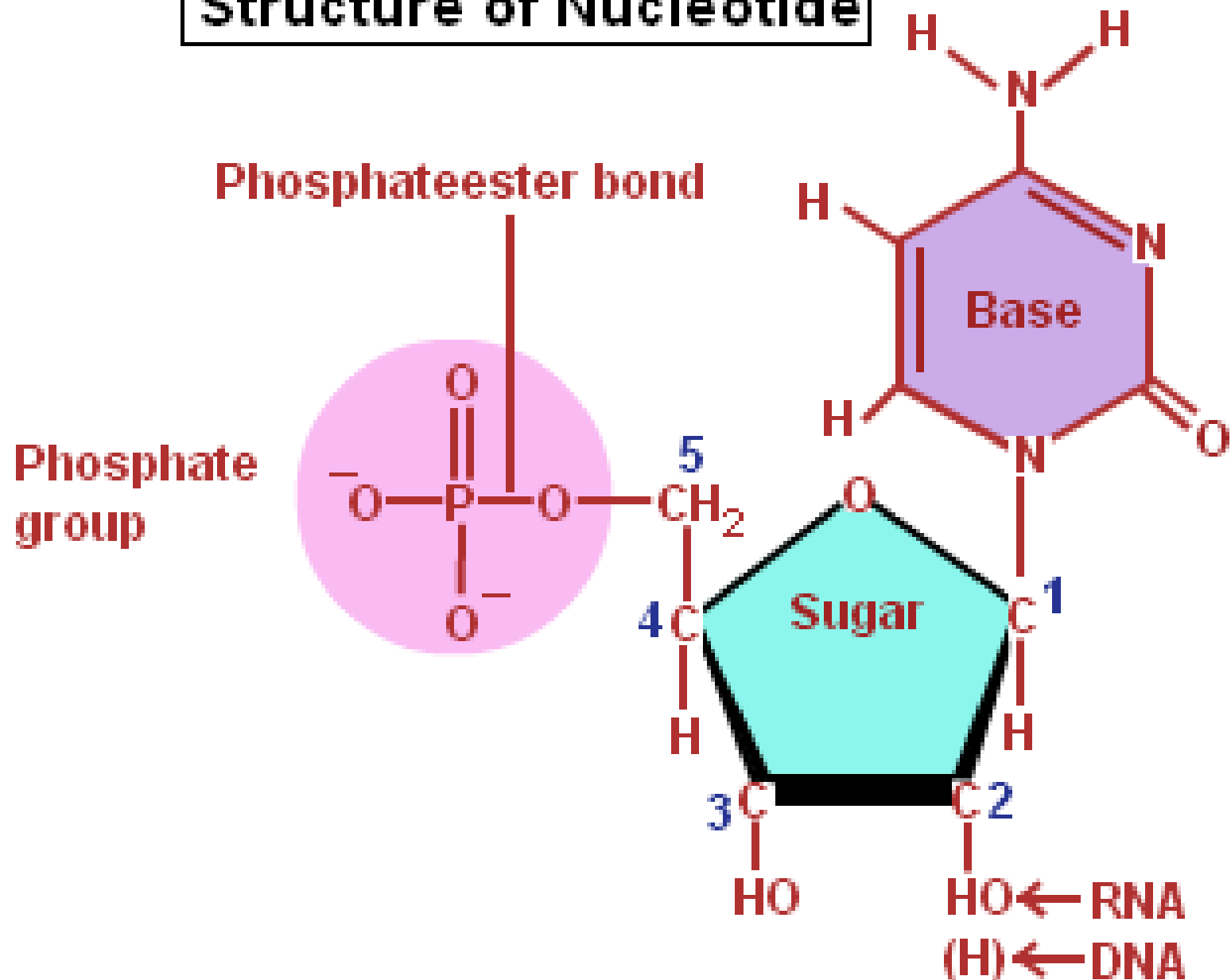


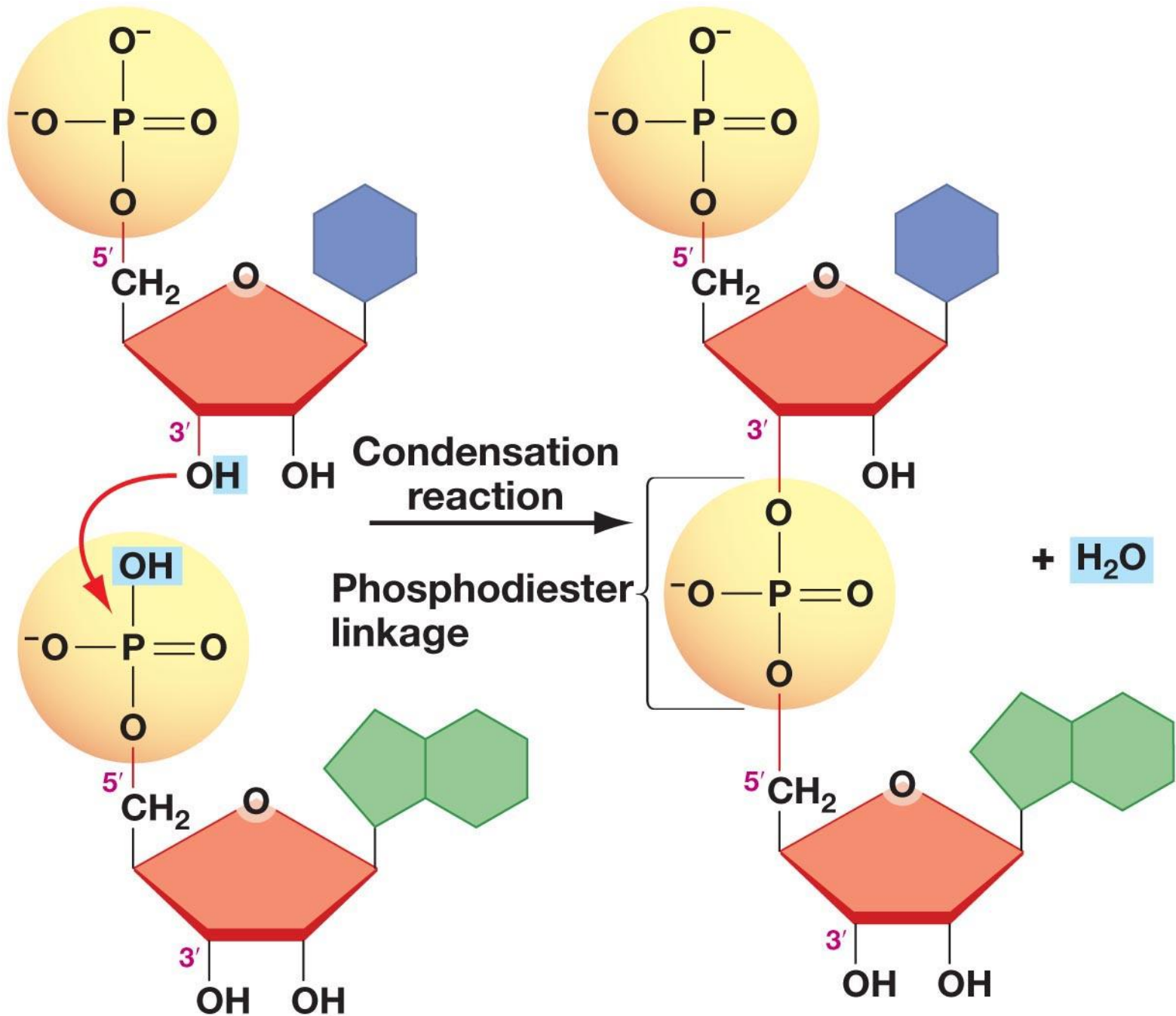
2-deoxy
guanosine

Nucleotides



Structure of Nucleotide

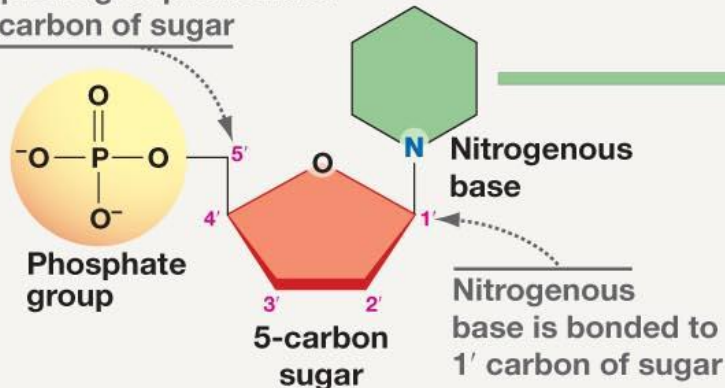




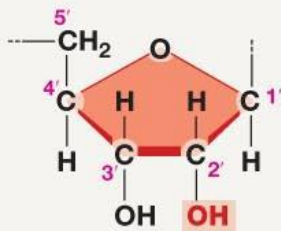
DNA contains thymine, but not uracil, while RNA contains uracil but not thymine

(a) Nucleotide

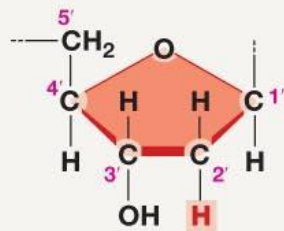
Phosphate group is bonded to 5' carbon of sugar



(b) Sugars

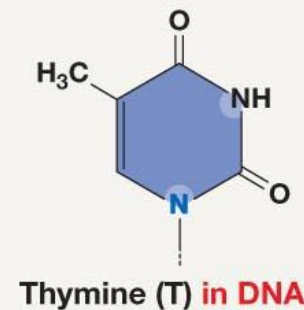
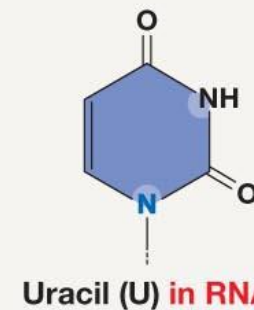
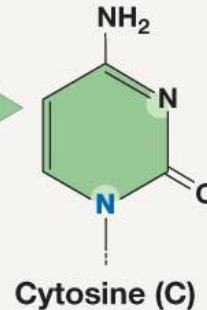


Ribose in RNA

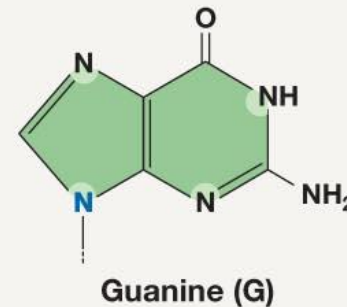


Deoxyribose in DNA

(c) Nitrogenous bases



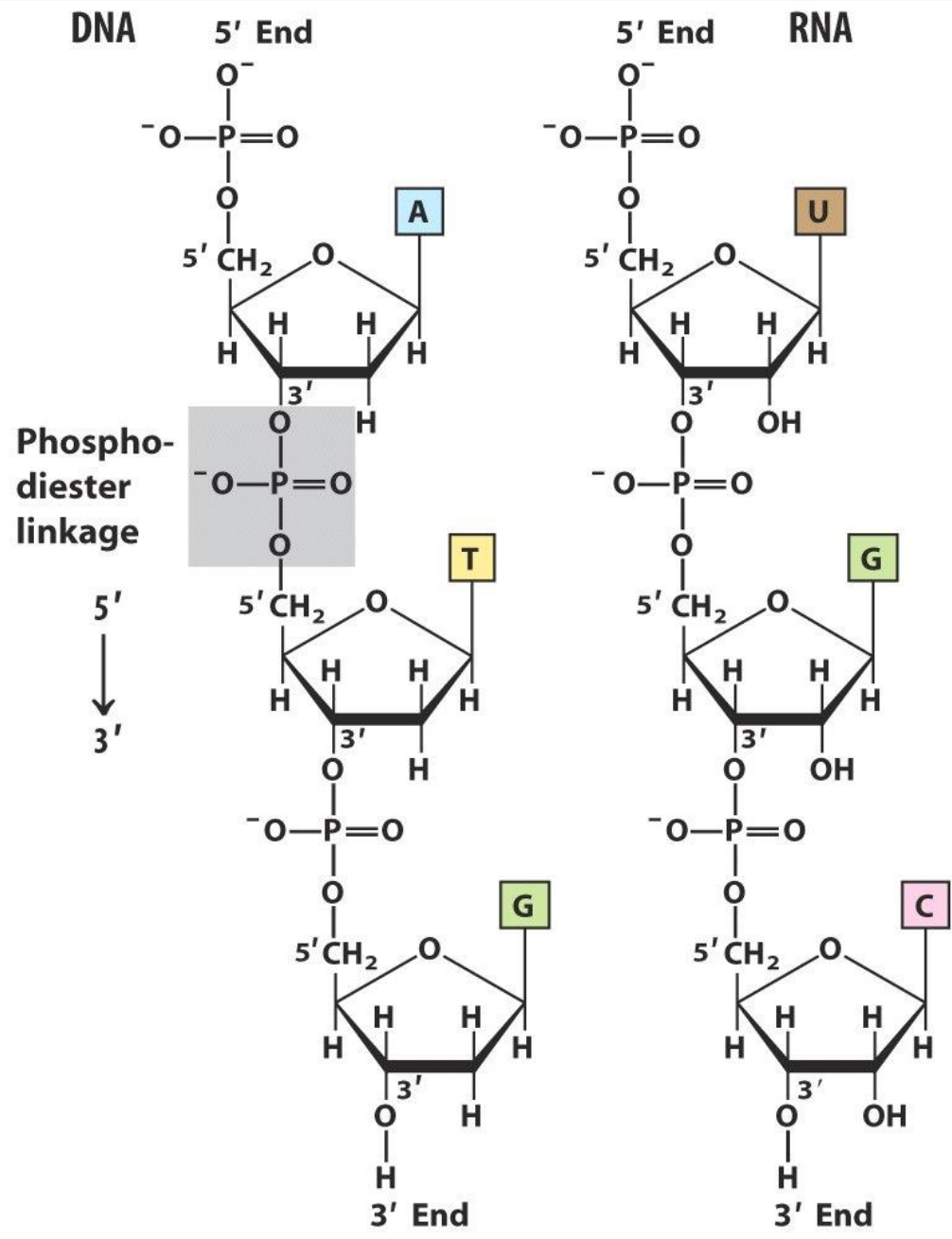
Pyrimidines



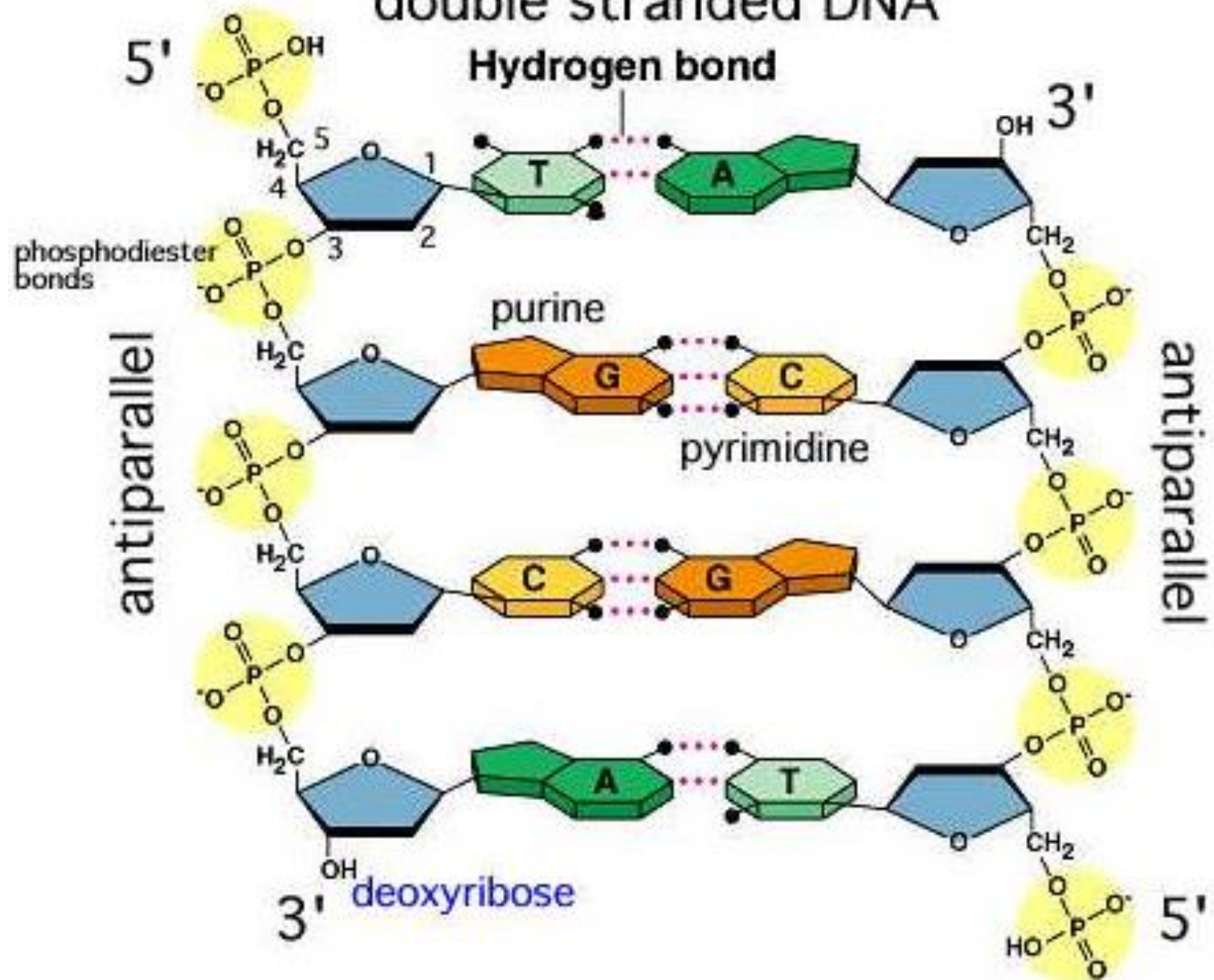
Purines are larger than pyrimidines

Purines

Polynucleotides



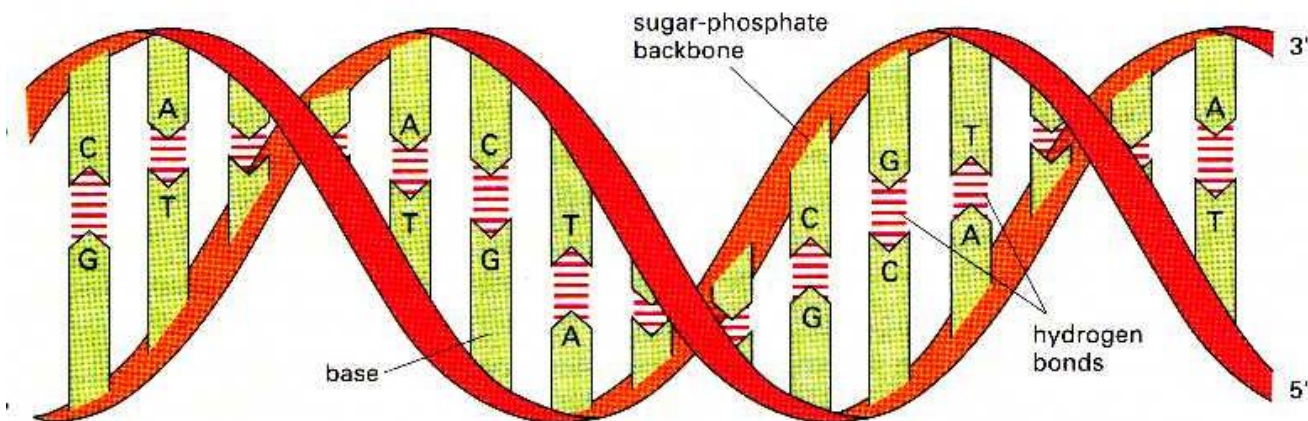
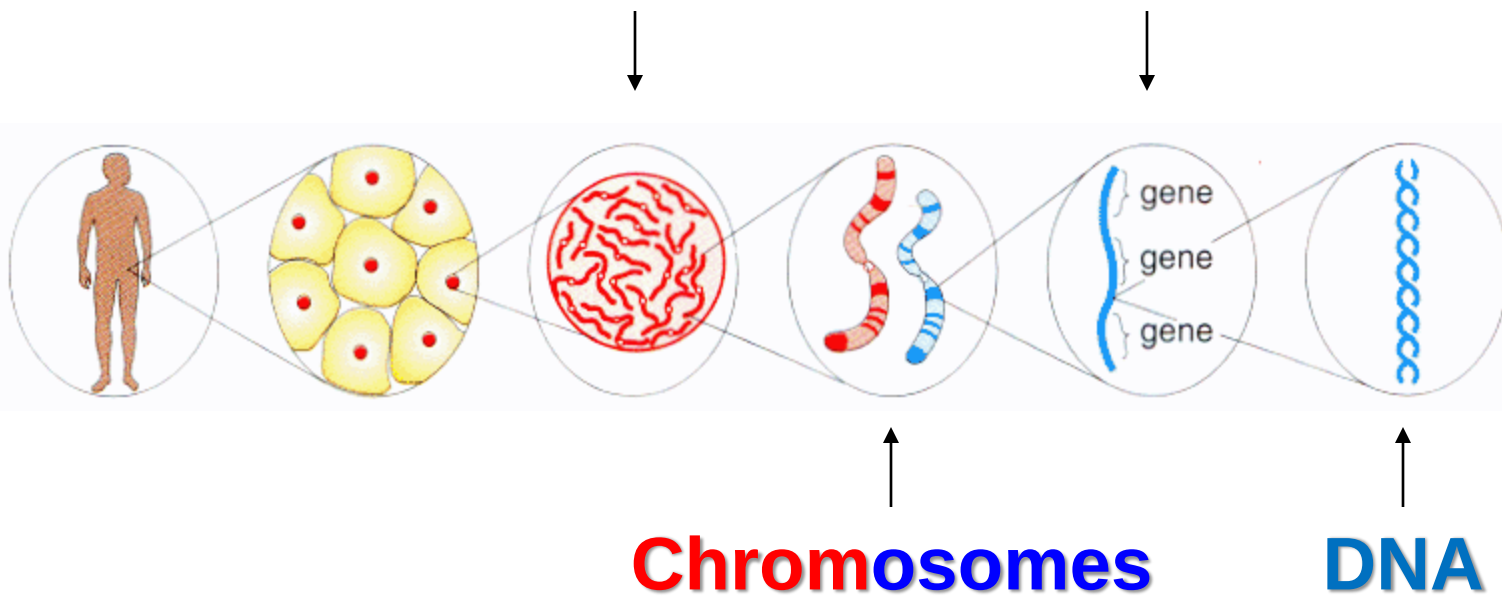
double stranded DNA



S. No.	DNA	RNA
1	Sugar moiety is Deoxy ribose	Sugar moiety is Ribose
2	The bases present are Adenine, Thymine, Guanine and Cytosine. Uracil is not present.	The bases present are Adenine, Uracil, Guanine and Cytosine. Thymine is rarely present.
3	Double stranded molecules	Single stranded molecules
4	Obeys Chargaff's rule	Does not obey Chargaff's rule
5	Bases are not modified	Bases are modified
6	It is stable and not hydrolysed easily by alkalis	It is unstable and hydrolysed easily by alkalis
7	DNA content is constant in all the cells except during cell division	Varies from cell to cell
8	The life time of DNA is comparatively high.	RNA is short lived.
9	No natural DNA is catalytic	RNA can be catalytic
10	Present in the nucleus, mitochondria and chloroplast	Present in the nucleus, mitochondria, nucleolus, ribosomes and cytosol.

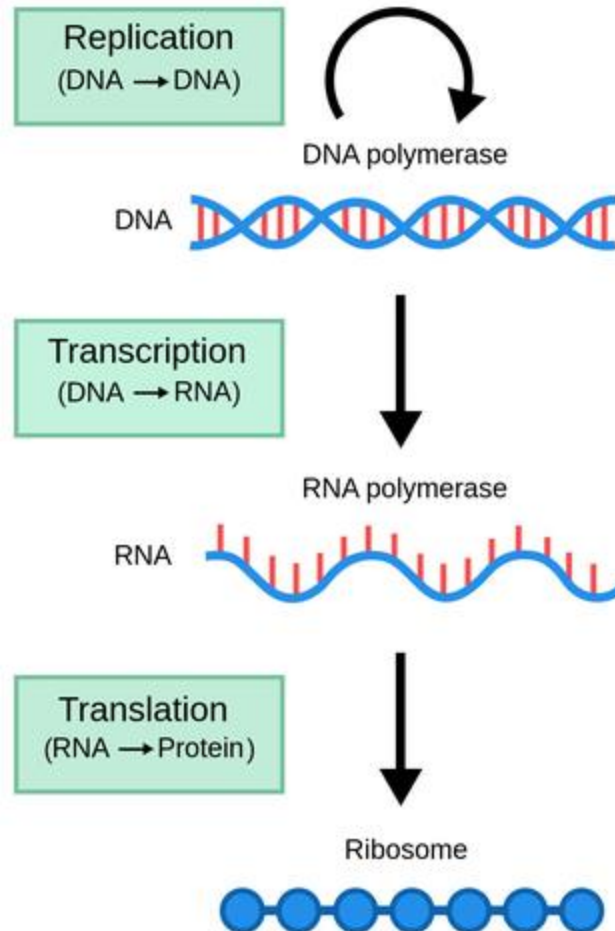
Cells

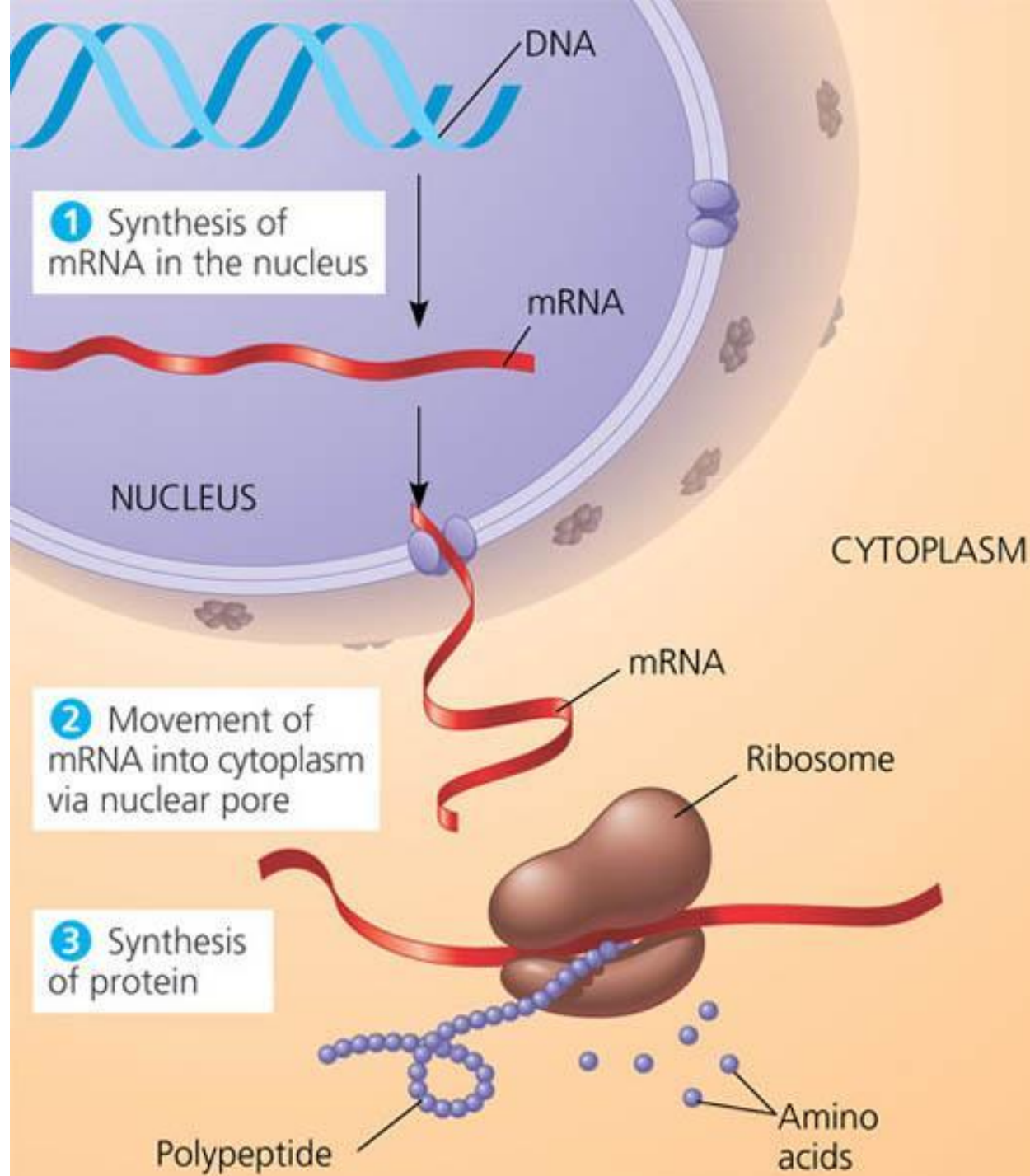
Genes



Central dogma of molecular biology

The central dogma of molecular biology is an explanation of the flow of genetic information within a biological system.





THANK YOU